



NUS
National University
of Singapore



Quantitative Reasoning with Data

Mr Samuel YEUN

Overview

- Introduction
- Population
- Research Question

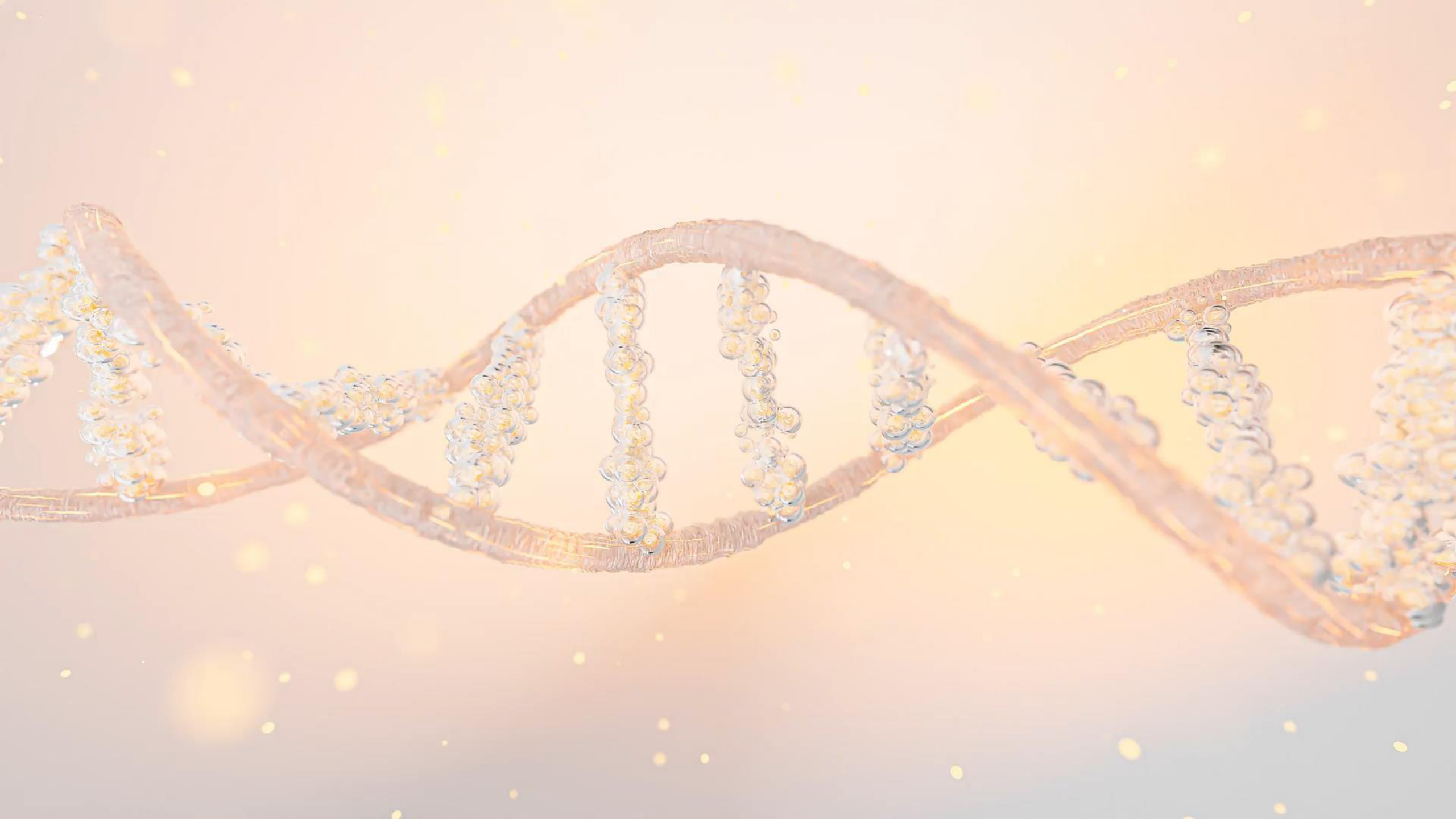


TDM	729.89	915.51	185.62▲25.43%	FLR	660.27	745.28	85.01▲12.88%
HUM	749.73	924.29	174.56▲23.28%	UVD	155.59	181.57	25.98▲16.70%
DMW	833.72	1004.01	170.29▲20.43%	QUV	440.55	540.21	99.66▲22.62%
YZJ	903.49	1127.46	223.97▲24.79%	HZT	285.51	344.98	59.47▲20.83%
GLY	982.07	1219.39	237.32▲24.17%	PCW	811.44	1029.66	218.22▲26.89%
VDA	113.74	143.41	29.67▲26.09%	AIK	361.77	451.39	89.62▲24.77%
UVV	468.08	535.41	67.33▲14.38%	ZJJ	858.36	994.57	136.21▲15.87%
HJS	545.49	659.05	113.56▲20.82%	RHJ	894.79	1046.68	151.89▲16.97%
EQC	566.96	664.69	97.73▲17.24%	VGV	425.08	509.95	84.87▲19.97%

PPJ	912.63	1038.36	125.73▲13.78%	ZBK	391.59	491.48	99.89▲25.51%
UAQ	1309.55	1655.62	346.07▲26.43%	BNY	969.21	1130.65	161.44▲16.66%
DAQ	1295.17	1641.66	346.49▲26.75%	SDM	735.44	913.39	177.95▲24.20%
PNR	654.33	775.84	121.51▲18.57%	TQJ	1323.91	1646.42	322.51▲24.36%
ZFM	751.53	909.13	157.6▲20.66%	QIS	543.42	667.24	123.82▲22.79%
				TTB	1455.17	1822.16	366.99▲25.24%







Introduction

Fall in Singapore marriages, divorces in 2020 amid Covid-19 restrictions, uncertainty

Source: Nabilah A., TODAY (2023)



Source: García P., Engaged couple linking fingers, 2017, Unsplash, <https://unsplash.com/photos/gc0BvNeGfuM>

The number of marriages in Singapore in 2020 fell 10.9 per cent from the year before, the Singapore Department of Statistics said.

Introduction



Number of marriages
in 2020:
22,651
(10.9% lower than in 2019)

"Covid-19 is to blame for the drop in marriages?"

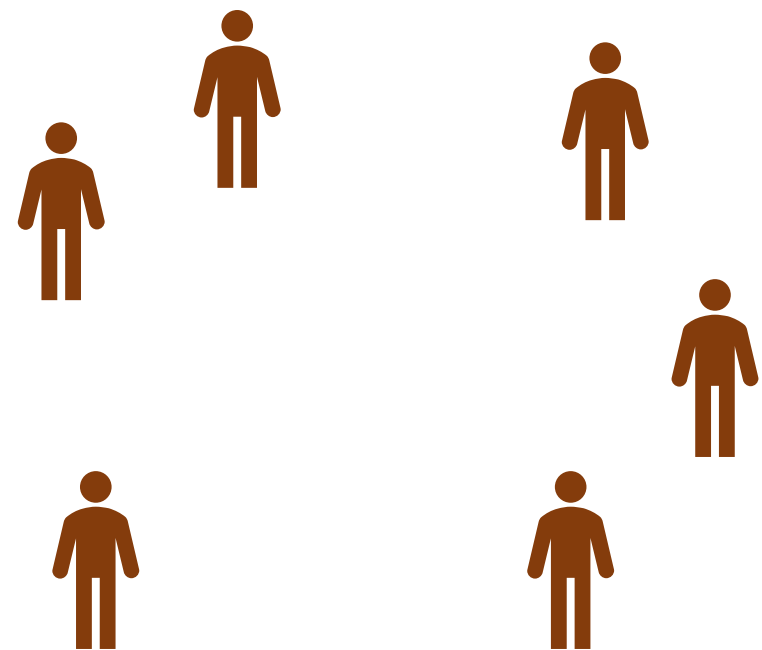
*Singapore's population size
for 2019 and 2020?*

*How did the study obtain
the number 22,651?*

Quantitative Reasoning with Data

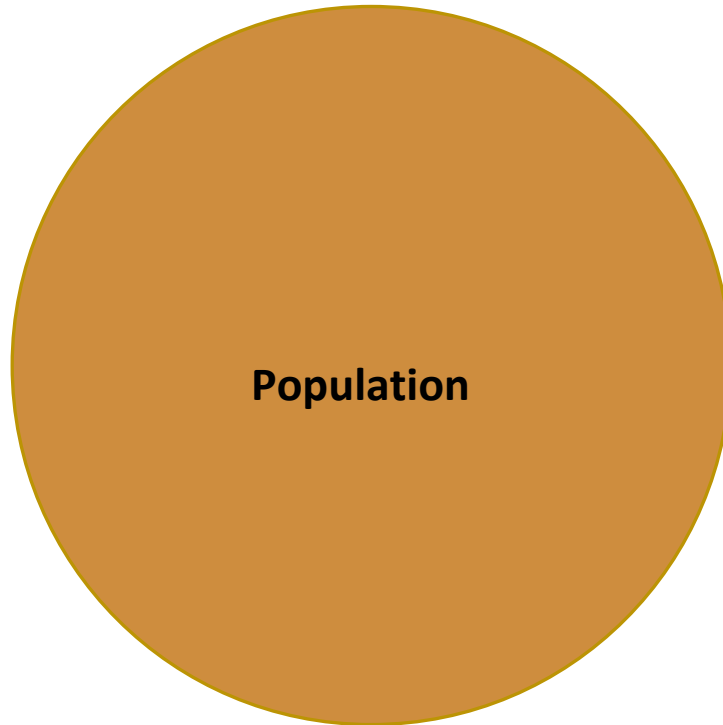
Population & Research Question

Population



Population

The population is the entire group (of individuals or objects) that we want to know something about.



Research Question

The research question is usually one that seeks to investigate some characteristic of a population.

- *What percentage of Singapore adults owns a car?*
(Population: Singapore adults)

- *Does Brand X pesticide work against mosquitoes?*
(Population: Mosquitoes)

- *Do NUS students that take notes using pen and paper score better than those using laptops?*
(Population: NUS students)

Research Question

Type of Research Questions	Examples
Make an estimate about the population	What is the average number of hours that students study each week?
	What proportion of all Singapore students is enrolled in a university?

Research Question

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	Does the majority of students qualify for student loans?

Research Question

Type of Research Questions	Examples
Make an estimate about the population	What is the average number of hours that students study each week?
	What proportion of all Singapore students is enrolled in a university?
Test a claim about the population	Is the average course load for a university student greater than 20 units?
	Does the majority of students qualify for student loans?
Compare two sub-populations	In university X, do female students have a higher GPA score than male students?
Investigate a relationship between two variables in the population	Are student athletes more likely than non-athletes to do final year projects?
	Is there a relationship between the average number of hours students spend each week on Facebook and their GPA?
	Does drinking coffee help students pass the math exam?

Summary

- Introduction
- Population
- Research Question

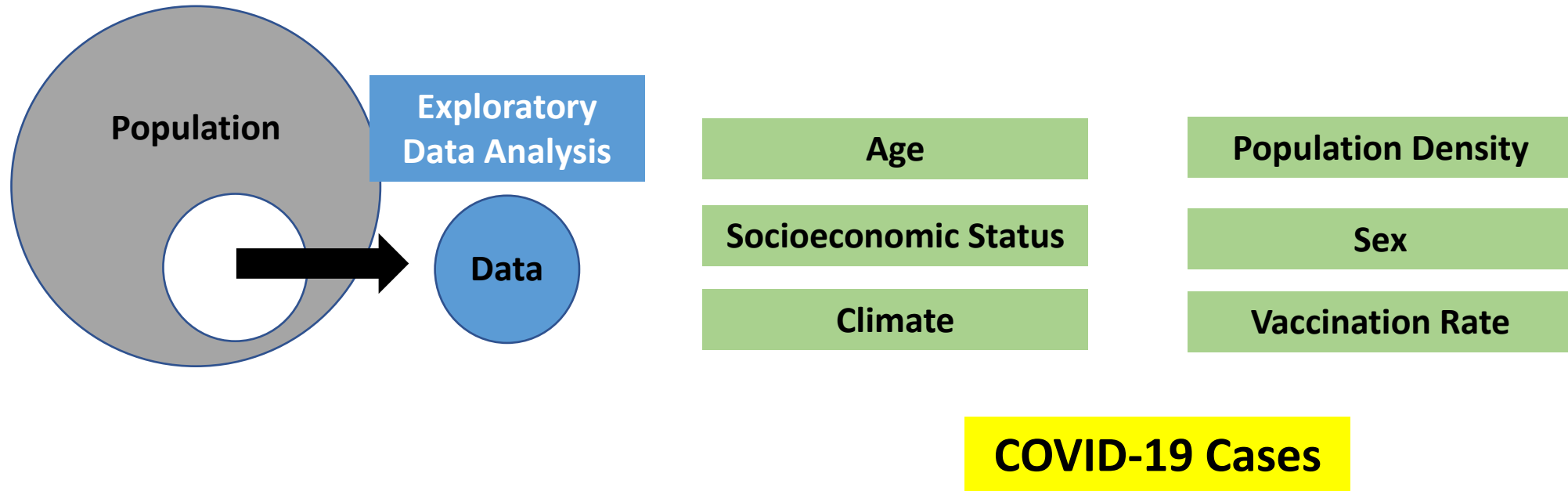
Introduction **to Sampling I**

Mr Timothy LIM

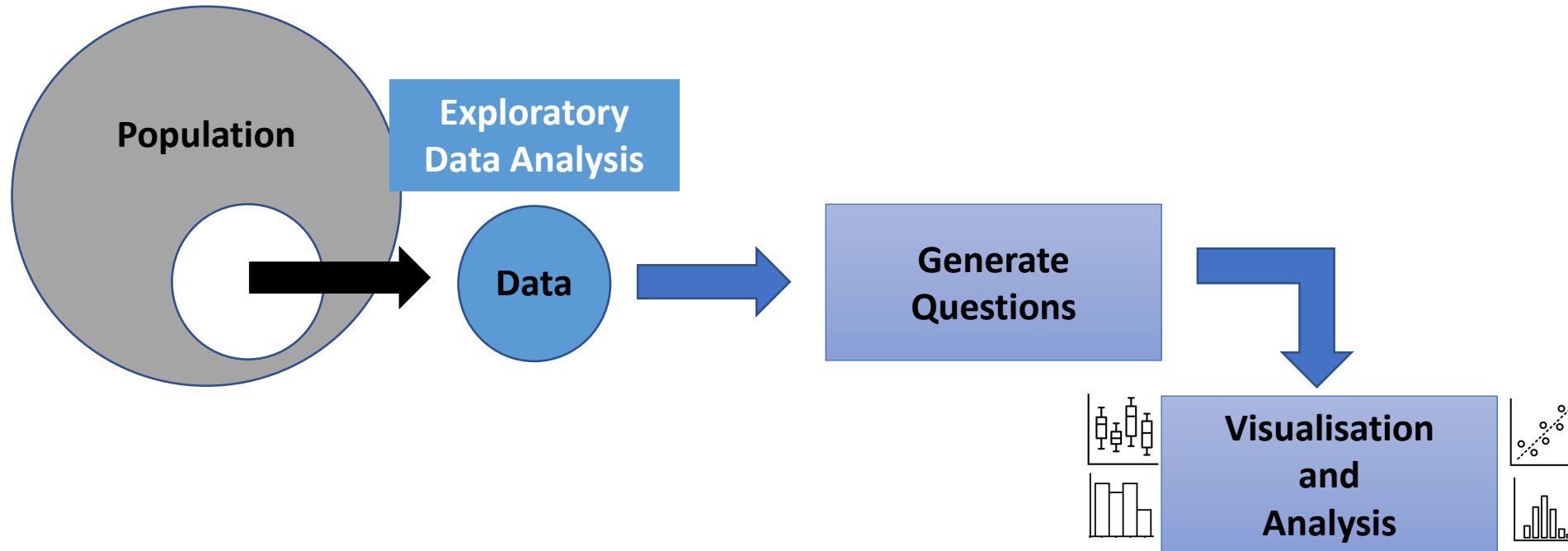
Overview

- Exploratory Data Analysis (EDA)
- Population, Sample and Sampling Frame
- Biases

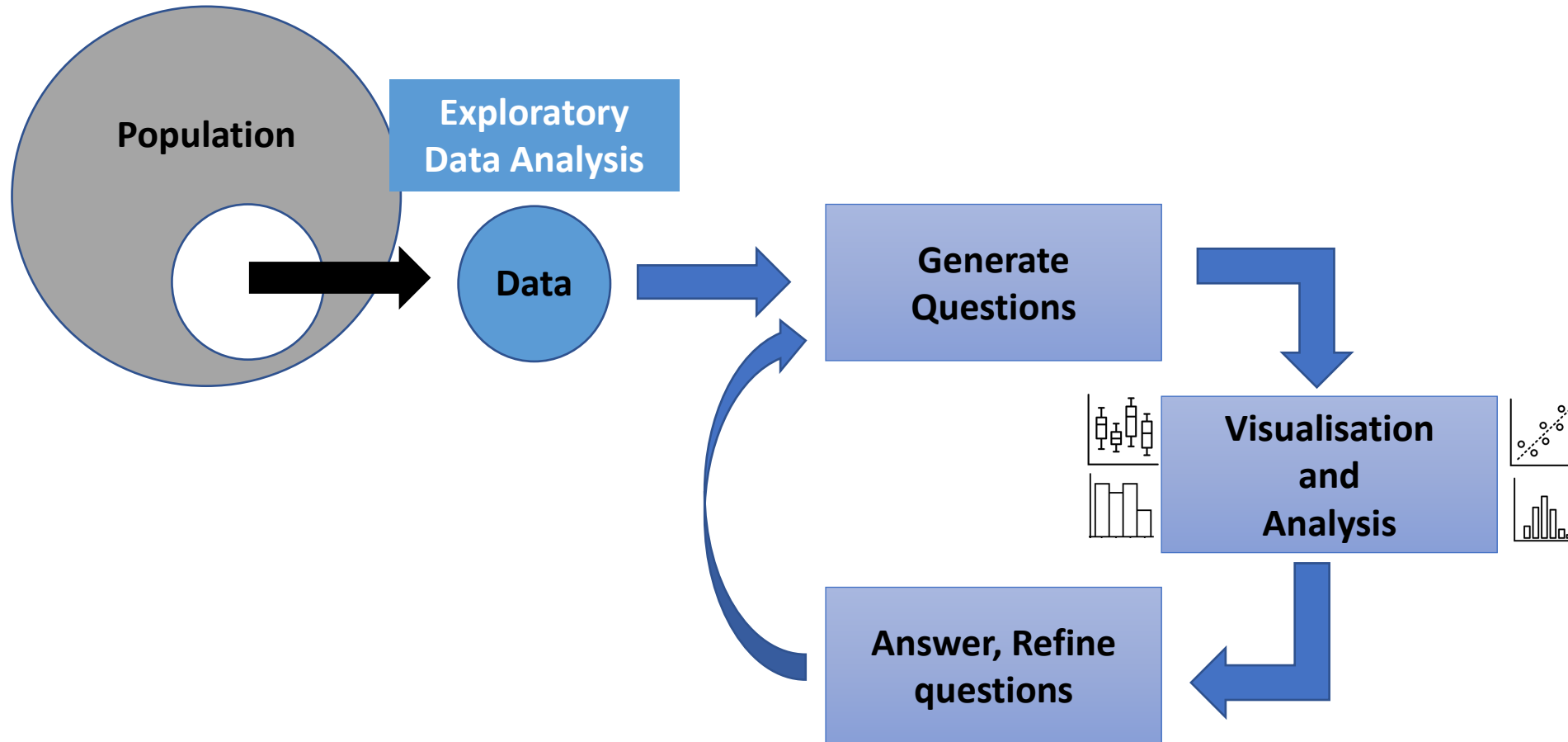
Exploratory Data Analysis



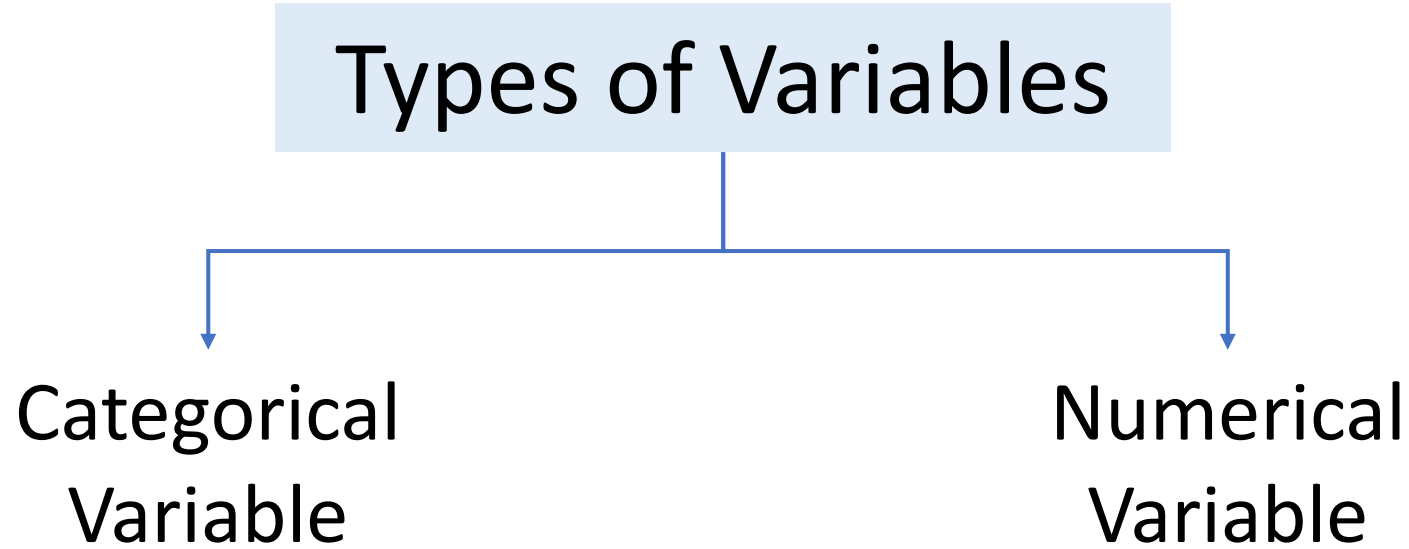
Exploratory Data Analysis



Exploratory Data Analysis



Types of Variables



```
graph TD; A[Types of Variables] --> B[Categorical Variable]; A --> C[Numerical Variable]
```

Categorical
Variable

Numerical
Variable

Types of Variables

```
graph TD; A[Types of Variables] --> B[Categorical Variable]; A --> C[Numerical Variable];
```

**Categorical
Variable**

Numerical
Variable

Types of Variables

```
graph TD; A[Types of Variables] --> B[Categorical Variable]; A --> C[Numerical Variable];
```

Categorical
Variable

**Numerical
Variable**

Types of Variables

```
graph TD; A[Types of Variables] --> B[Categorical Variable]; A --> C[Numerical Variable]; B --> D[Association between variables];
```

Categorical Variable

Numerical Variable

Association
between variables

Types of Variables

```
graph TD; A[Types of Variables] --> B[Categorical Variable]; A --> C[Numerical Variable]; B --> D[Association between variables]; C --> E[Correlation]; style C fill:#d9e1f2
```

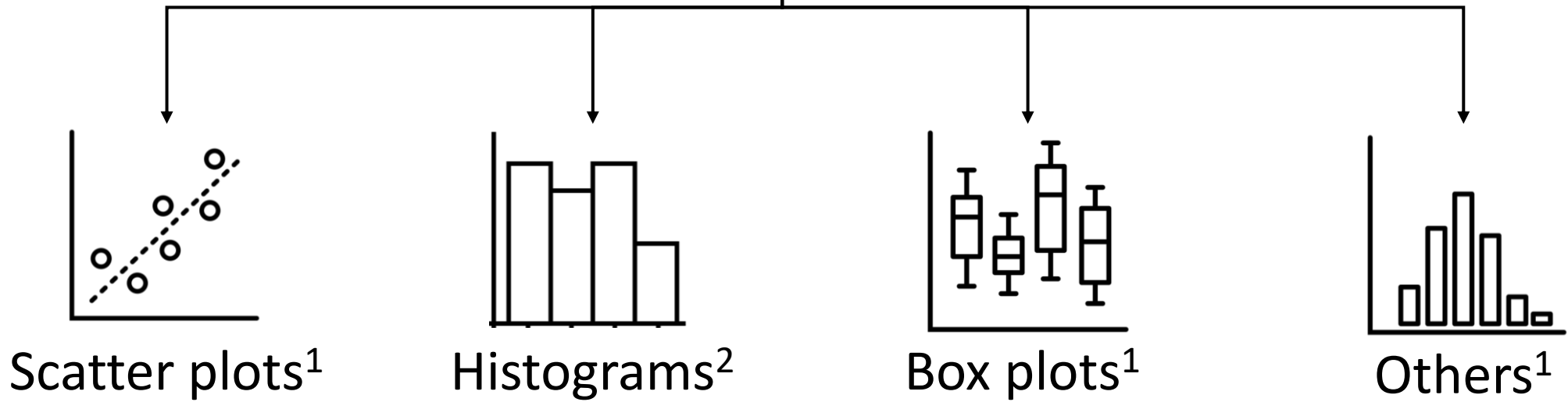
Categorical Variable

Association
between variables

Numerical Variable

Correlation

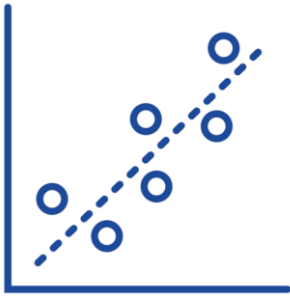
Visualisation Charts



¹ Source: White E., <https://thenounproject.com/browse/collection-icon/data-visualization-32102/?p=1>

² Source: Agra S., <https://thenounproject.com/icon/histogram-2312373/>

Visualisation Charts



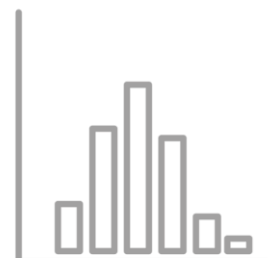
Scatter plots



Histograms

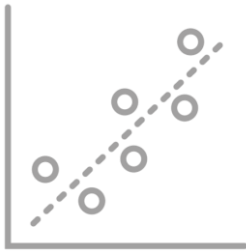


Box plots

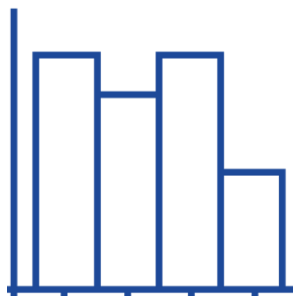


Others

Visualisation Charts



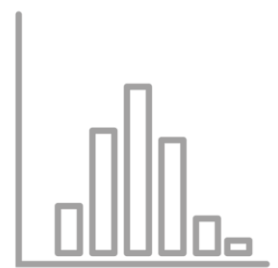
Scatter plots



Histograms



Box plots



Others

Visualisation Charts



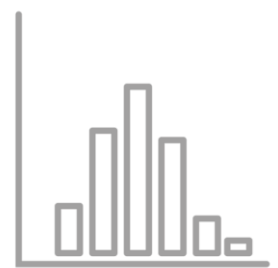
Scatter plots



Histograms

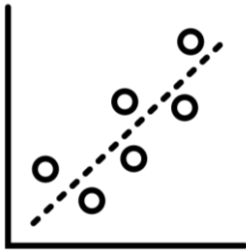


Box plots

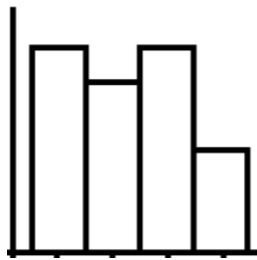


Others

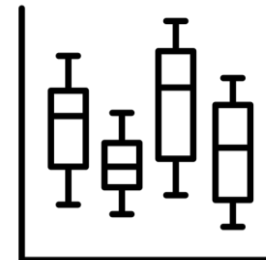
Visualisation Charts



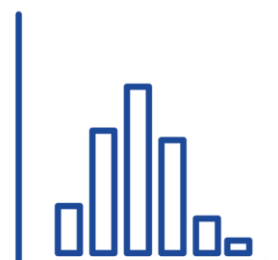
Scatter plots



Histograms

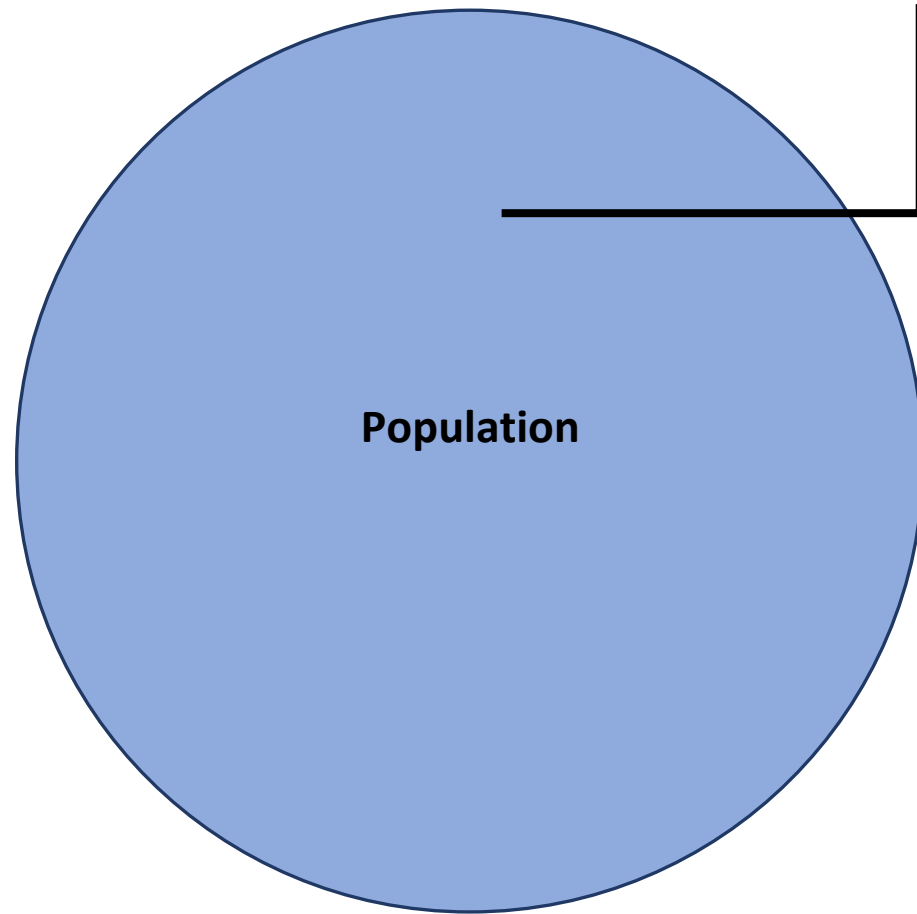


Box plots



Others

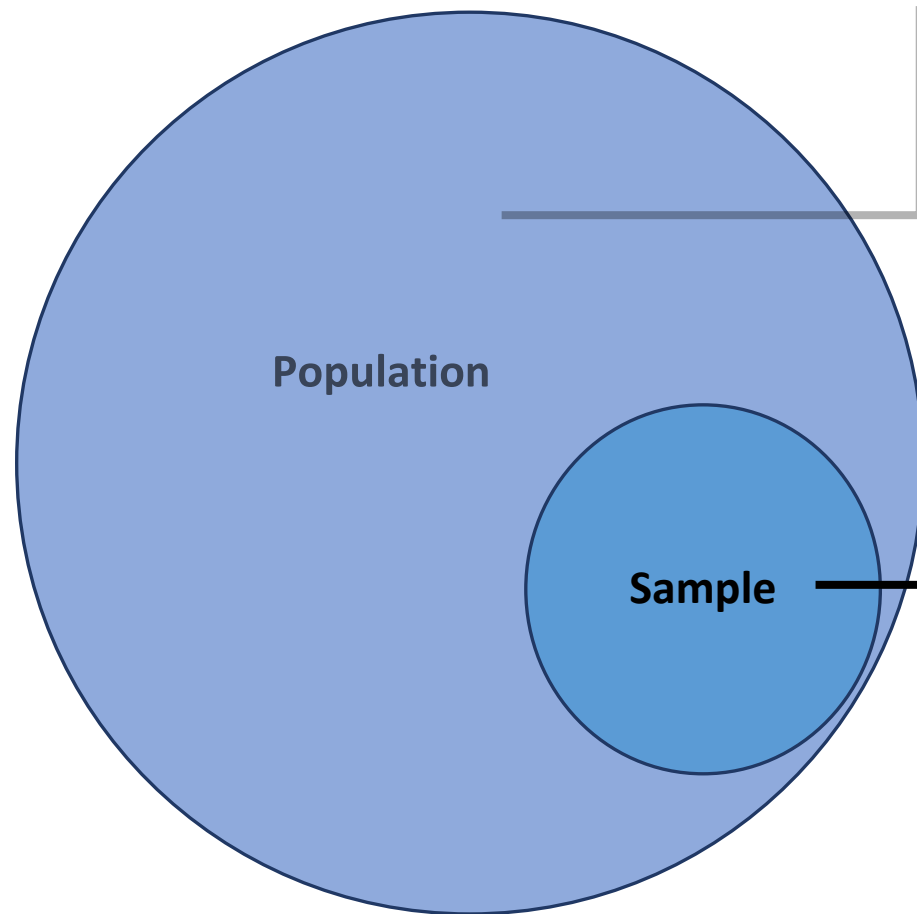
Population vs Sample



Population of Interest

- A group in which researcher has interest in drawing conclusions of the study.
 - **Population Parameter** – A numerical fact about a population.

Population vs Sample



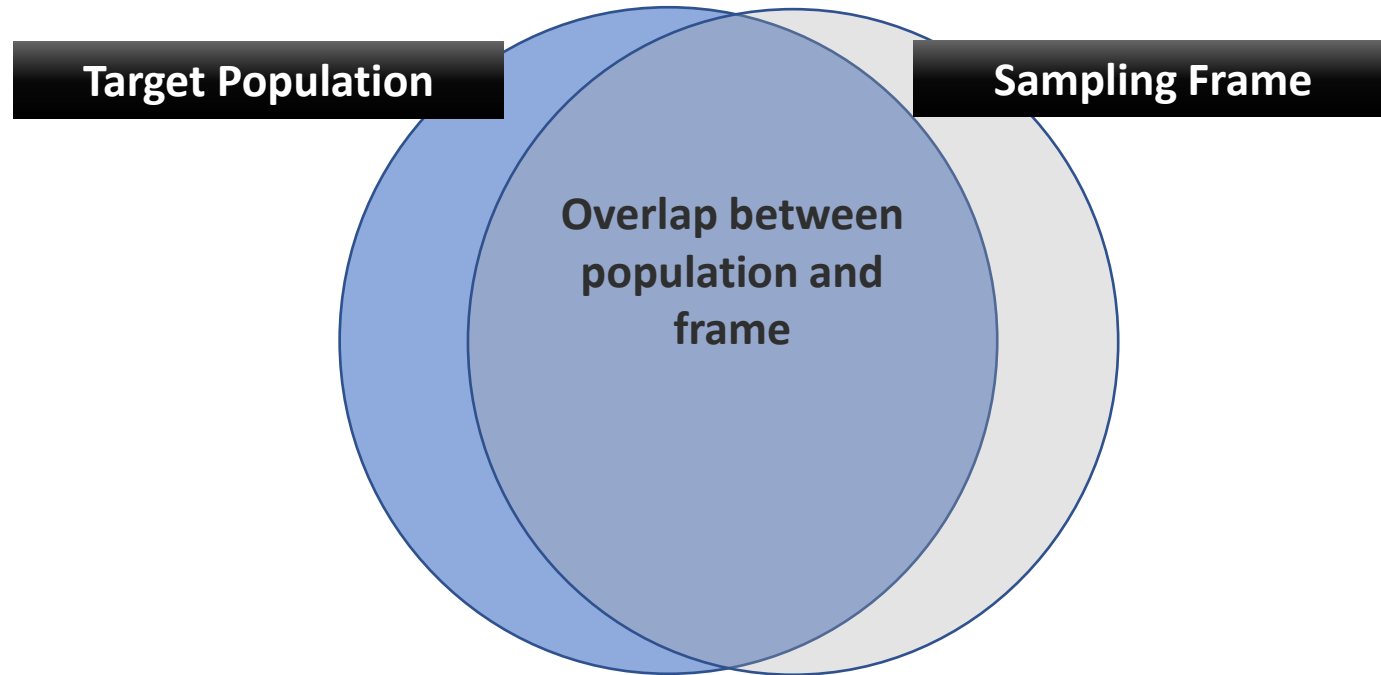
Population of Interest

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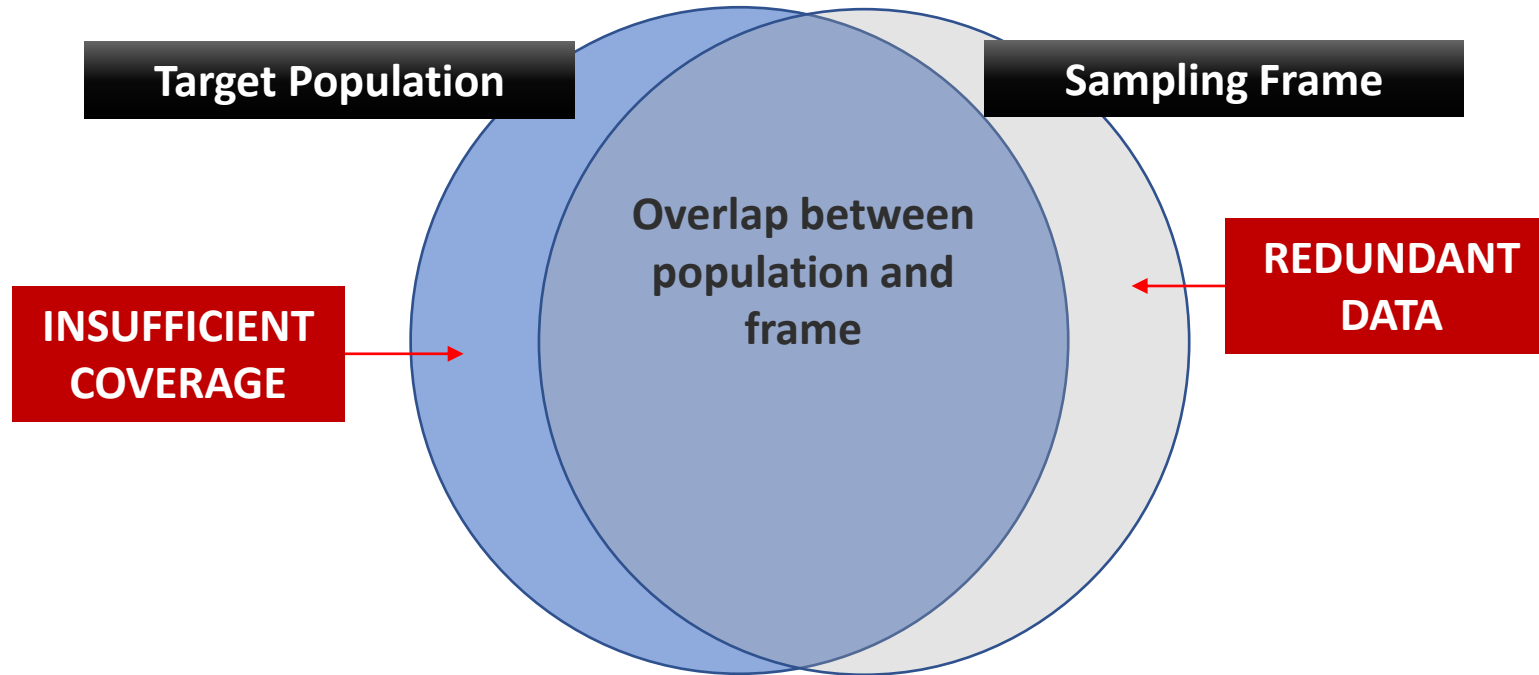
Sample

- A proportion of the population selected in the study.
 - **Estimate** – An inference based on information from a sample.

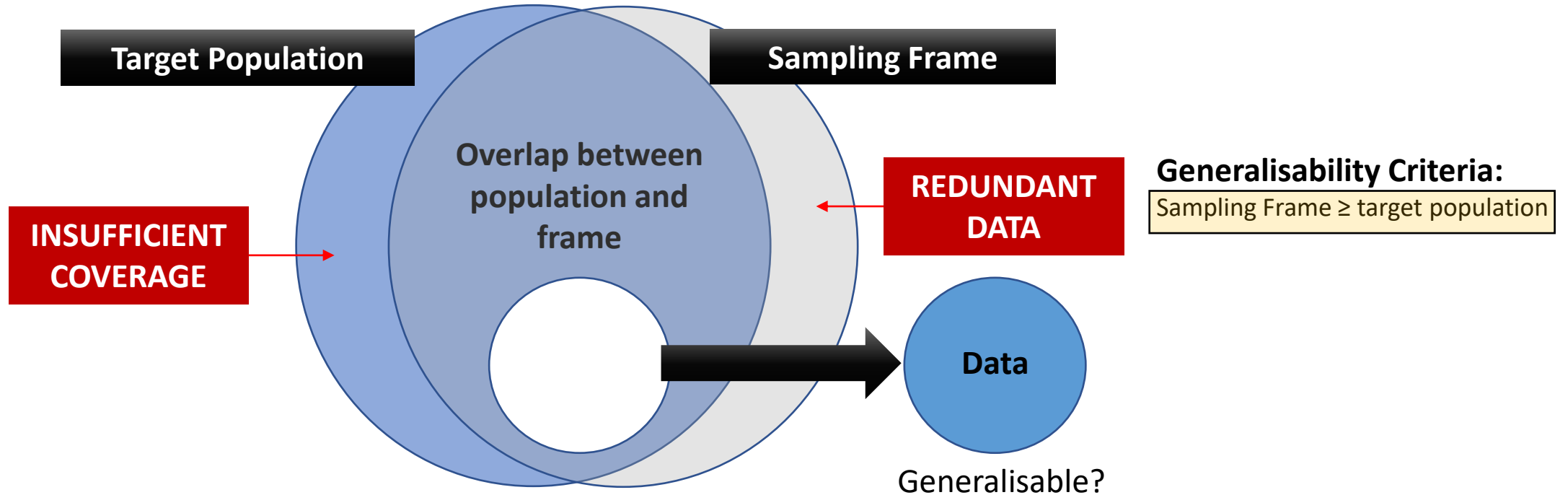
Sampling Frame



Sampling Frame



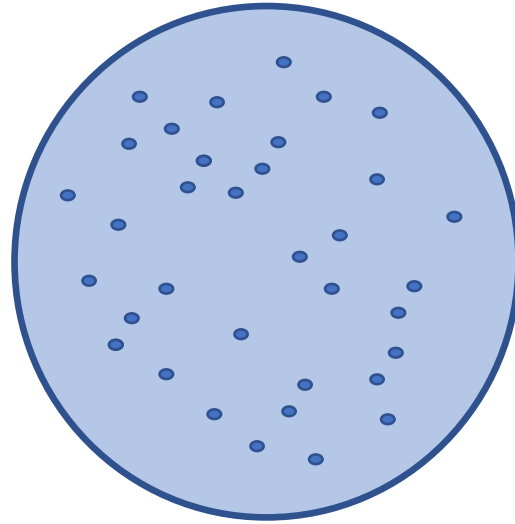
Sampling Frame



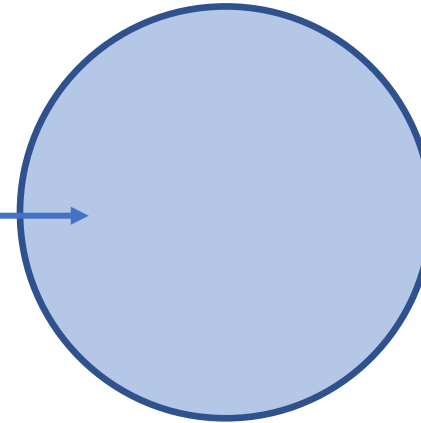
Census vs Sample

Census – An attempt to reach out to the whole population

Population



Sample



Sample – A proportion of population selected

- Cost
- Speed

Watch out for **BIASES**

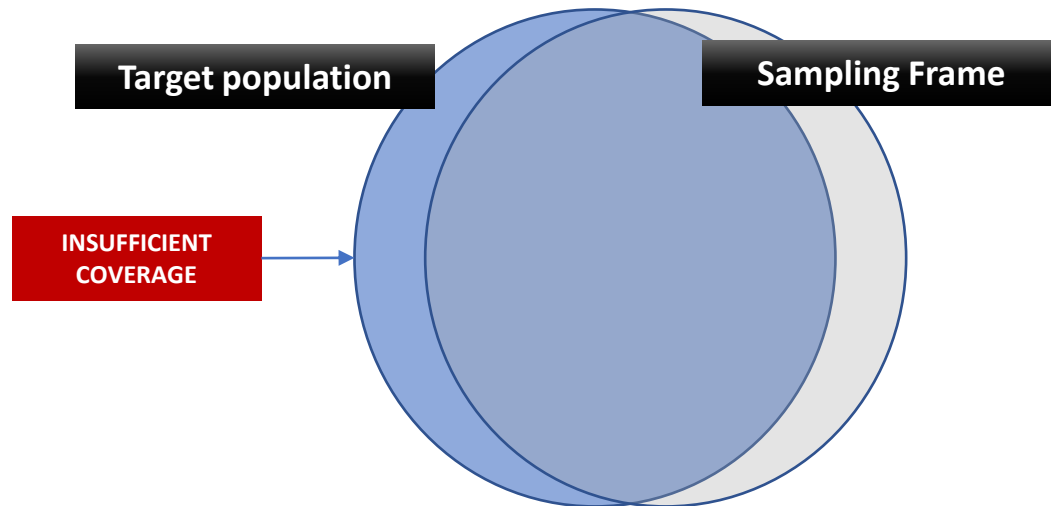
Selection Bias

Non-response Bias

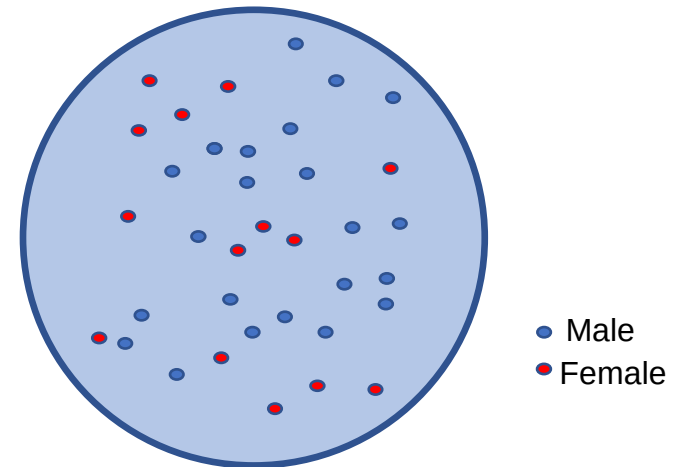
Selection Bias

Biased selection of units

Imperfect Sampling Frame

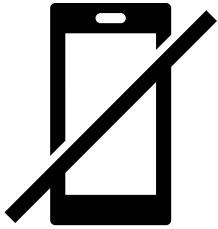


Non-Probability Sampling



Non-Response Bias

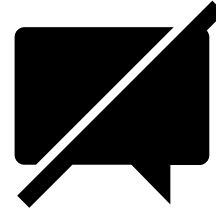
Non-disclosure of information



Not Interested



Inconvenient



Information
Too Sensitive

A Prelude

- Sampling Methods
 - ✓ Probability Sampling
 - ✓ Non-Probability Sampling
- Generalisability Criteria



Introduction **to Sampling II**

Mr Timothy LIM

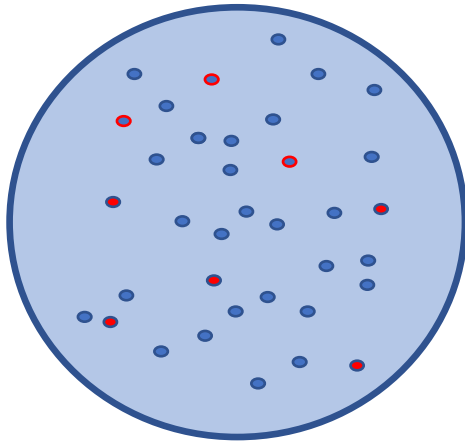
Overview

- Probability Sampling
- Non-Probability Sampling
- Generalisability

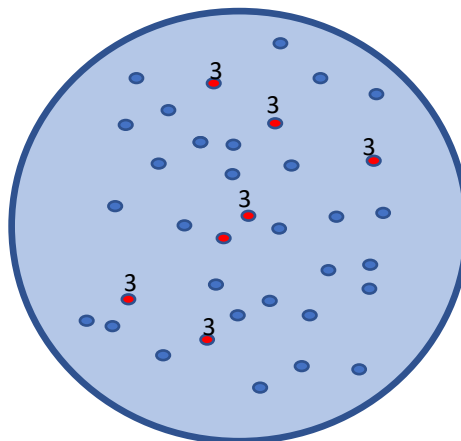
What is Probability Sampling?

- Sampling process is via a **known** randomised mechanism
- The probability of selection **may not be the same** throughout all units of sampling frame
- Element of **chance** in selection process **eliminates biases** associated with selection

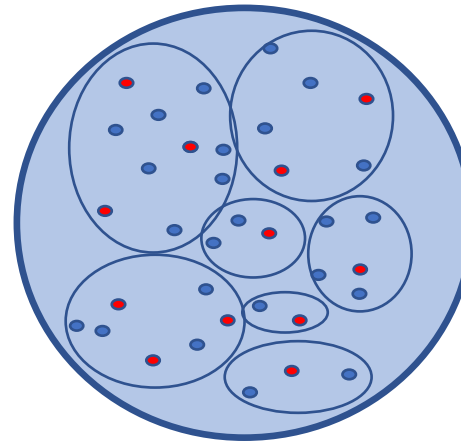
Simple Random



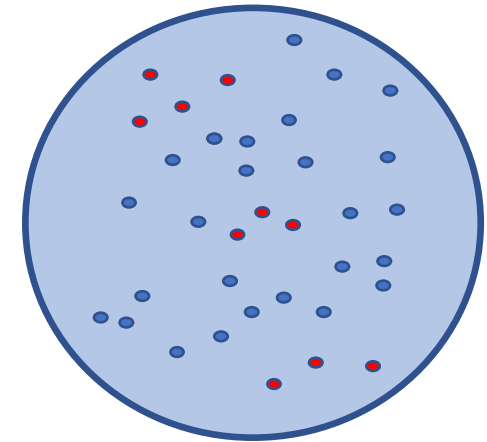
Systematic



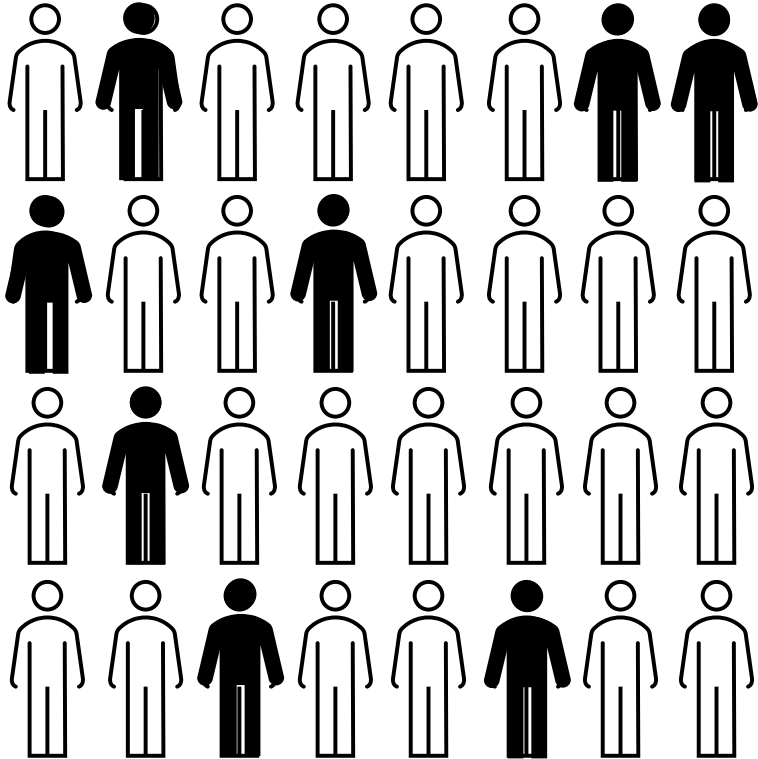
Stratified



Cluster



Simple Random Sampling

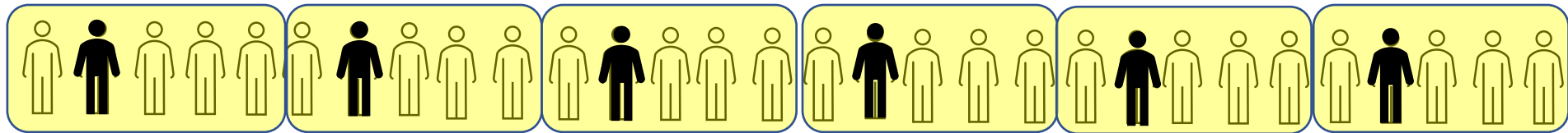


- Units are selected randomly **without replacement** from the sampling frame
- Mechanism: Random Number Generator
- Every set of units has equal chance to be the sample actually selected
- Results do not change haphazardly between samples as variability is due to chance

Advantage	Good Representation
Disadvantage	Non-response

Systematic Sampling

- Applying a selection interval K , and *random starting point from first interval*.



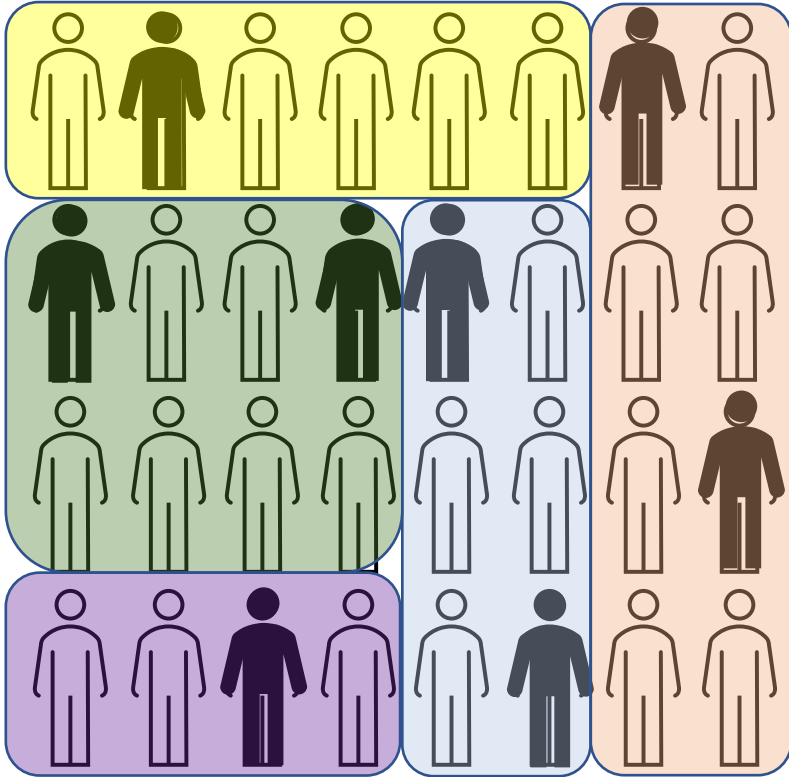
Systematic Sampling

- Applying a selection interval K , and random starting point from first interval.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70

Advantage	Simpler Process
Disadvantage	Not good representation

Stratified Random Sampling



- Population broken down into strata of different size
- Units within strata share similar characteristics, but characteristics vary across different strata
- Simple Random Sampling is done within every stratum

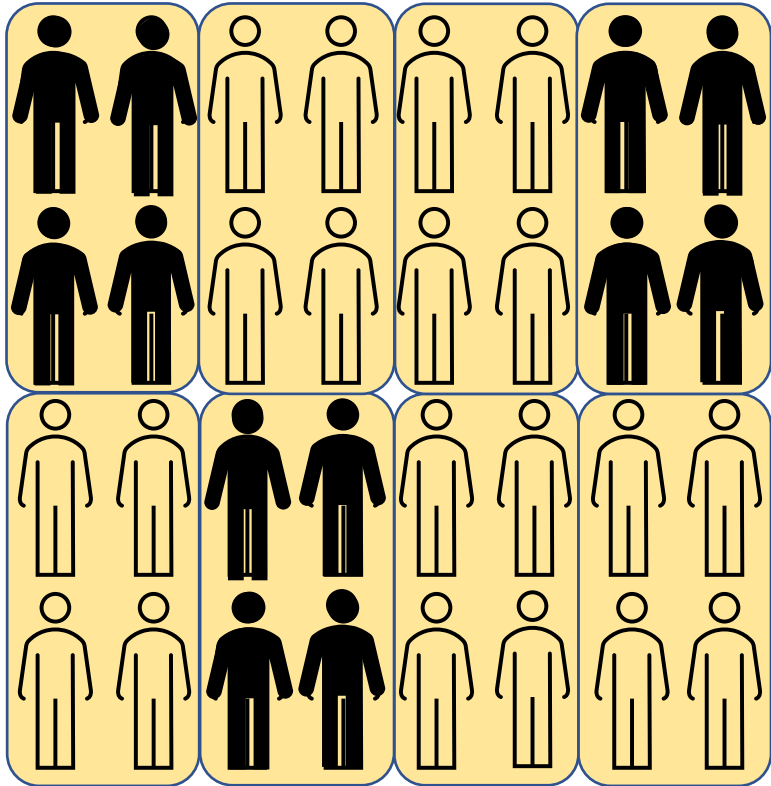
Advantage

Good Representation

Disadvantage

Need information about Sampling Frame and Strata

Cluster Sampling



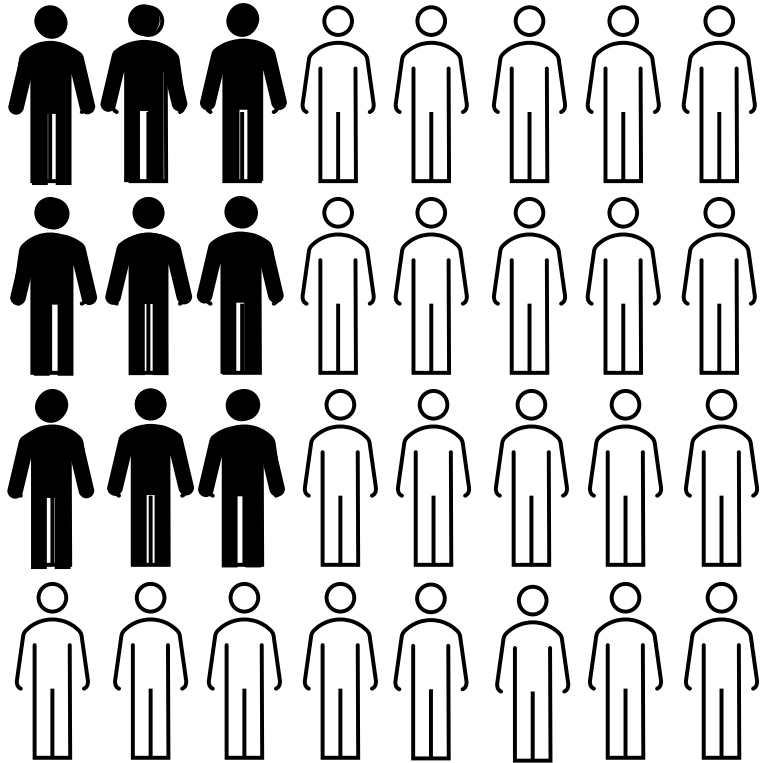
- Units broken down into **similar** clusters
- Randomly sample **fixed** number of **clusters**
- All **observations within selected clusters** are recorded.

Advantage	Less tedious, time-consuming, costly
Disadvantage	High variability if clusters are dissimilar

Pros and Cons of Probability Sampling

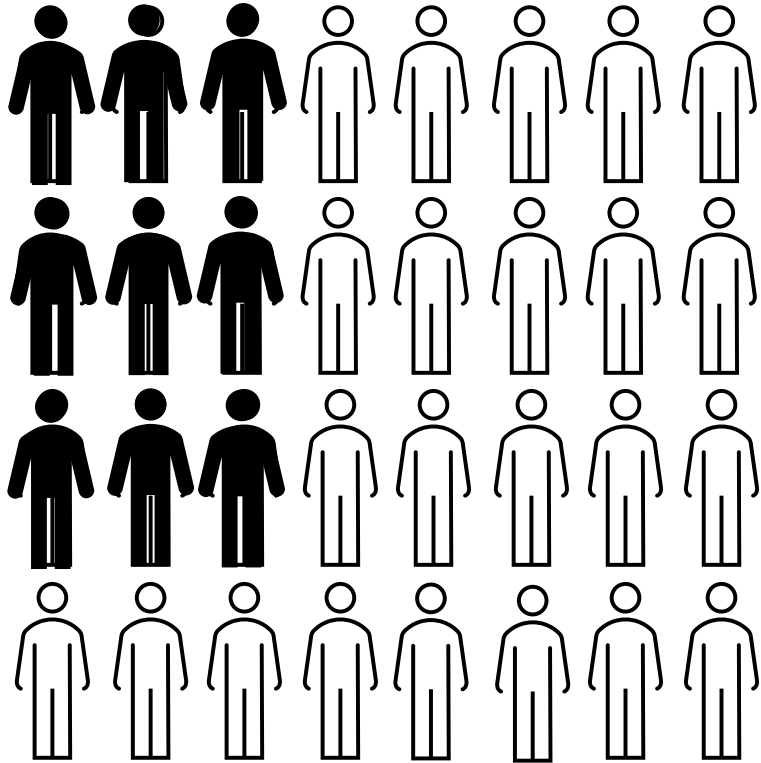
Sampling Plan	Advantages	Disadvantages
Simple Random Sample	Good Representation of the Population	Time-consuming; accessibility of information
Systematic Sample	Simpler selection process as opposed to Simple Random Sampling	Potentially under-representing the population
Stratified Random Sample	Good Representation of Sample by Stratum	Require Sampling Frame and criteria for classification of population into stratum
Cluster Random Sample	Less time-consuming and less costly	Require larger sample size in order to achieve low margin of error

What is Non-Probability Sampling?



- The non-usage of chance in selection of individuals
- Examples include: Quota Sampling, Convenience Sampling, Judgment Sampling, Volunteer Sampling
- Sampling methods are not mutually exclusive

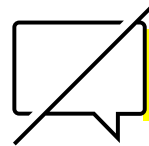
Example 1: Convenience Sampling



- Selection of subjects based on proximity and availability
- Case Example: Mall Surveys

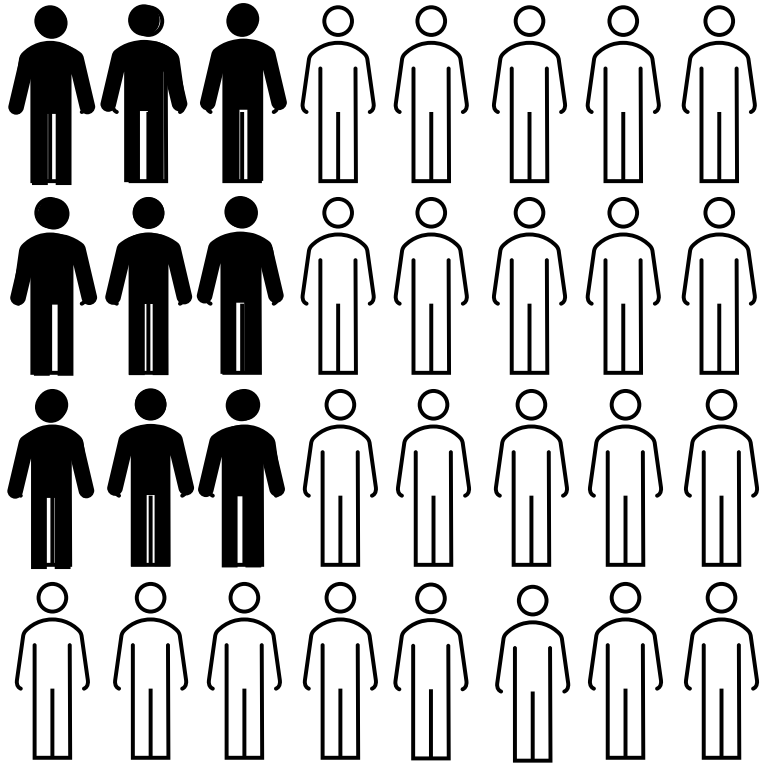


Selection Bias



Non-Response Bias

Example 2: Volunteer Sampling



- Non-Probability sampling method in which researchers actively seek volunteers to participate
- Case Example: Online Polls (American Family Association, 2004)



Selection Bias?

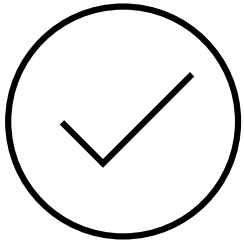


Non-Response Bias?

General Approaches to Sampling

- Choose Appropriate Sampling Frame
- Use a Probability-Sampling Method
- Remove Unwanted Units

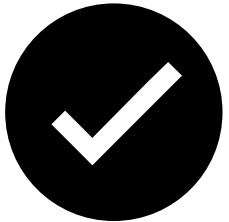
The Criteria for Generalisability



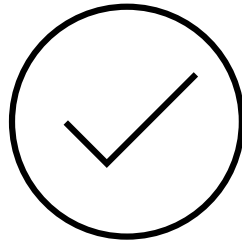
**Good
Sampling Frame**



**Probability-based
Sampling**



**Large
Sample Size**



**Minimum
Non-response**



Variables and Summary Statistics

Variables and Means

Mr Sudarshan NARASIMHAN

Overview

- Types of variables
 - Definition of a variable
 - Independent and dependent variables
 - Types of variables
- Means
 - The reason for summary statistics
 - Definition and properties of mean
 - Overall means versus subgroup means
 - Proportion as a form of mean

What is a variable?

A **variable** is an attribute that can be measured or labelled

Race

Age

Population Density

Socioeconomic Status

Time

Sex

A data set consists of individuals and variables pertaining to the individuals

A	B	C	D	E	F	G
Title	Production Budget	Worldwide Gross	Domestic Gross	Date of Release	CPI	Duration
Star Wars Ep. VII: The Force Awakens	306000000	2064615817	936662225	16/12/2015	237	2h 18m
Spider-Man: No Way Home	200000000	1910041582	814108407	14/12/2021	271	2h 28m

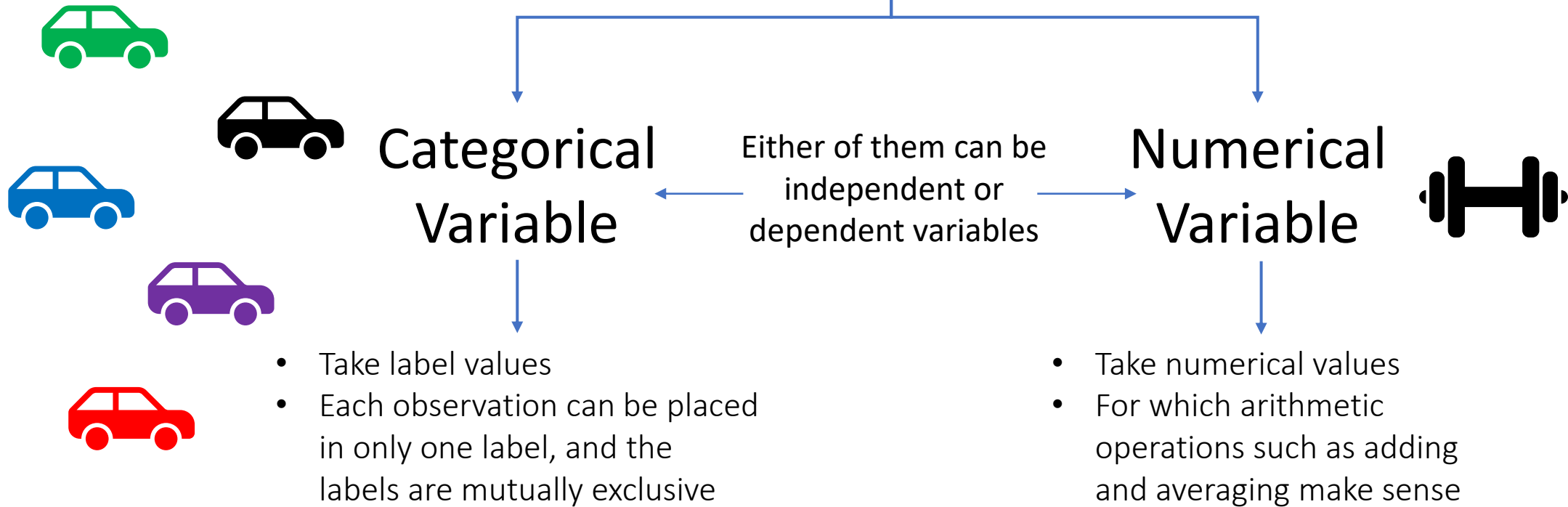
Independent and dependent variables

An **independent variable** is a variable that may be subject to manipulation (either deliberately or spontaneously) in a study.

A **dependent variable** is a variable which is *hypothesised to change* depending on how the independent variable is manipulated in a study.

Research question	Dependent/Independent variable
Does amount of caffeine consumed per day affect the quality of sleep amongst Singaporean adults?	Independent variable : Amount of caffeine consumed per day
	Dependent variable : Quality of sleep

Types of Variables





Categorical variables

Ordinal

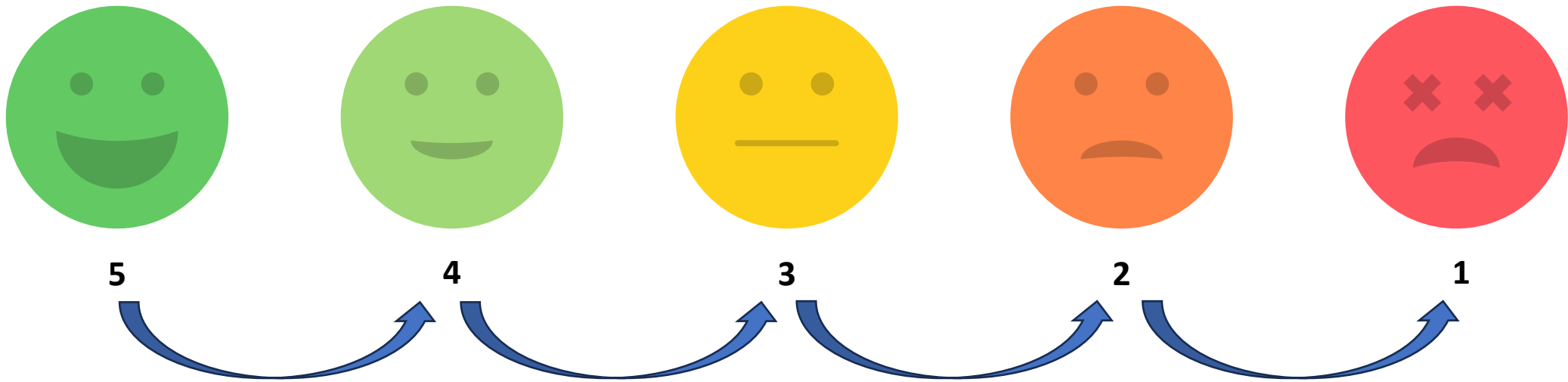
Comes with some natural ordering and numbers are often used to represent the ordering.

Nominal

In other cases where there is no intrinsic ordering for the variables.



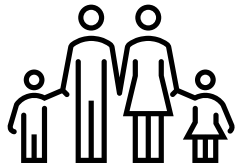
WARNING



These differences are subjective and therefore not consistent
Labelling categories using numbers does not transform the nature of the variable
to become numerical.

Numerical variables

What does it mean
to have 3.75
members??!!



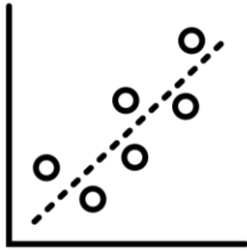
Discrete

Is one where possible values
of the variable form a set of
numbers with “gaps” .

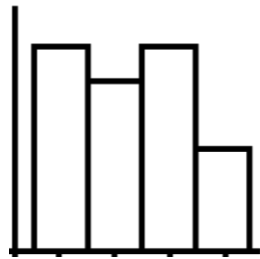
Continuous

Is one that can
meaningfully take on all
possible numerical values
in a given range or
interval.

Why bother classifying variables?



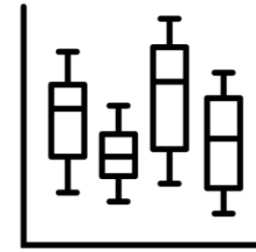
Scatter plots¹



Histograms²



Bar Graphs¹



Box plots¹

You need to know your variable types before deciding what visualisation tools to use

¹ Source: White E., *Data Visualization*, 2017, The Noun Project. CC0

² Source: Agra S., *histogram*, 2019, The Noun Project. CC0

What do I want to do
with my data?

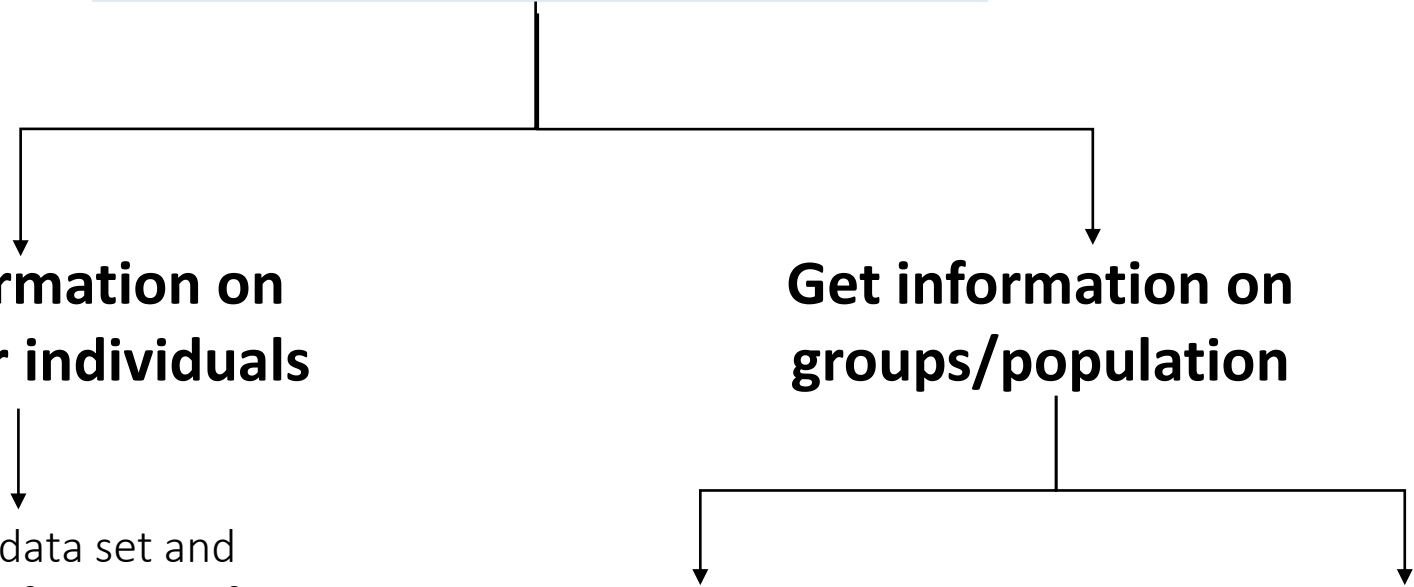
**Get information on
particular individuals**

Go to the data set and
extract the information for
the particular individual(s)

**Get information on
groups/population**

Data Visualisation

Summary Statistics



Summary Statistics

```
graph TD; A[Summary Statistics] --> B[Measures of Central Tendency]; A --> C[Measures of Dispersion]; B --> D["• Mean<br>• Median<br>• Mode"]; C --> E["• Standard Deviation<br>• Interquartile Range"]
```

Measures of Central Tendency

- Mean
- Median
- Mode

Measures of Dispersion

- Standard Deviation
- Interquartile Range

Mean

The **mean** of a numerical variable x , denoted by $\bar{x}$, (read “x bar”) in a data set is given by the formula

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

where n denotes the number of data-points and x_1 to x_n denote the values of the numerical variable x in the data.

The formula is also also written as $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

Case	Age	Gender	Race	Days to Recover
1	66	M	Chinese	26
2	53	F	Chinese	14
3	37	M	Chinese	27
4	36	M	Chinese	17
5	56	F	Indian	21
6	56	M	Malay	23
7	35	M	Malay	7
8	56	F	Chinese	20
9	56	M	Chinese	25
10	56	M	Chinese	10
11	31	F	Others	11
12	37	F	Indian	13

Properties of the mean

- $x_1 + x_2 + \dots + x_n = n\bar{x}$.
- Adding a constant value to all the data points changes the mean by that constant value.
- Multiplying all the values to all the data points by a constant number c will result in the mean also being multiplied by c .

Whether the weather be fine

	A	B	C	D	E
1	Station ▾	Year ▾	Month ▾	Day ▾	Daily Rainfall Total (mm) ▾
2	Changi	2020	January	1	0
3	Changi	2020	January	2	0
4	Changi	2020	January	3	0
5	Changi	2020	January	4	0
6	Changi	2020	January	5	0
7	Changi	2020	January	6	11
8	Changi	2020	January	7	4.8
9	Changi	2020	January	8	0
10	Changi	2020	January	9	0
11	Changi	2020	January	10	2.4
12	Changi	2020	January	11	0
13	Changi	2020	January	12	0
14	Changi	2020	January	13	0
15	Changi	2020	January	14	0
16	Changi	2020	January	15	0
17	Changi	2020	January	16	0
18	Changi	2020	January	17	0
19	Changi	2020	January	18	0
20	Changi	2020	January	19	0
21	Changi	2020	January	20	0

Source: Meteorological Service Singapore, *Historical Daily Records*, CCO
<http://www.weather.gov.sg/climate-historical-daily/>.

- Which months were the
 - Hottest?



- Coolest?



- Most wet?



Aggregating the data

	A	B	C	D	E
1	Station ▾	Year ▾	Month ▾	Day ▾	Daily Rainfall Total (mm) ▾
2	Changi	2020	January	1	0
3	Changi	2020	January	2	0
4	Changi	2020	January	3	0
5	Changi	2020	January	4	0
6	Changi	2020	January	5	0
7	Changi	2020	January	6	11
8	Changi	2020	January	7	4.8
9	Changi	2020	January	8	0
10	Changi	2020	January	9	0
11	Changi	2020	January	10	2.4
12	Changi	2020	January	11	0
13	Changi	2020	January	12	0
14	Changi	2020	January	13	0
15	Changi	2020	January	14	0
16	Changi	2020	January	15	0
17	Changi	2020	January	16	0
18	Changi	2020	January	17	0
19	Changi	2020	January	18	0
20	Changi	2020	January	19	0
21	Changi	2020	January	20	0

After aggregating the data, we can see that May and December were the wettest months



We can calculate the average rainfall per month which works out to be 157.22mm. But what does this average tell us?

month	total_rainfall
2020-01	88.4
2020-02	65
2020-03	108.8
2020-04	188
2020-05	255.6
2020-06	233.8
2020-07	140.8
2020-08	103.4
2020-09	150.2
2020-10	78.8
2020-11	220.6
2020-12	253.2

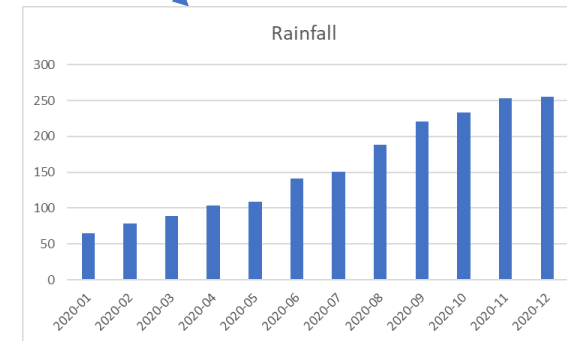
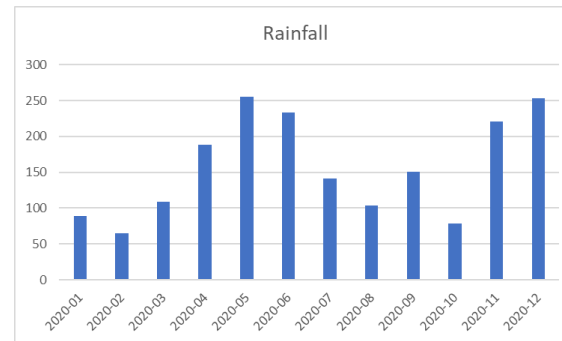
What the mean can/cannot tell us



Mean rainfall = 157.22mm/month

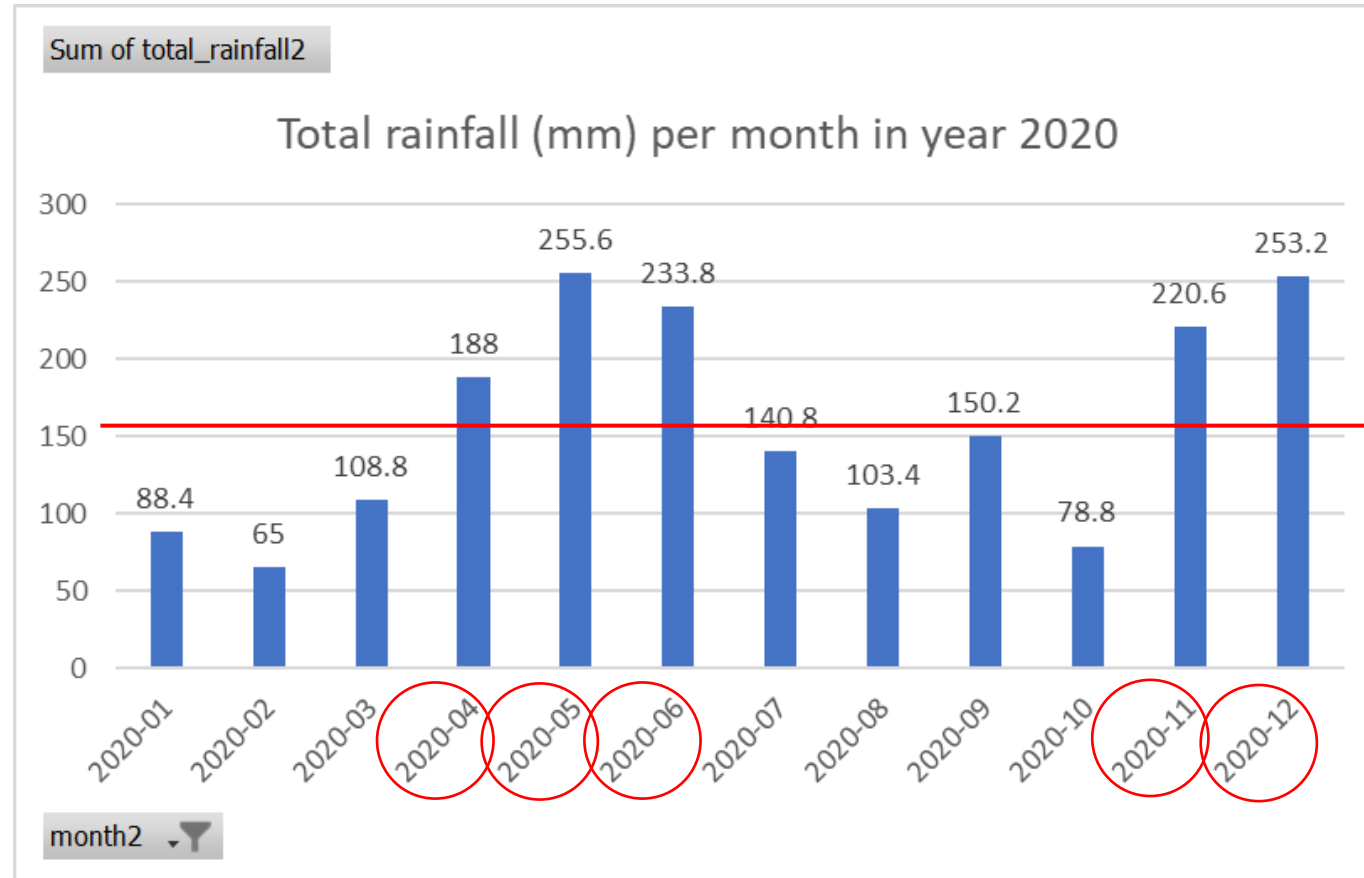
?

Total rainfall for 2020 :
 $157.22 \times 12 = 1886.6\text{mm}$



Both these distributions of rainfall have the same total and mean

A common misconception about the mean



Mean = 157.22mm

Overall vs subgroup means

	A	B	C	D	E
1	school ▼	Total ▼	sex ▼	age ▼	location ▼
2	School A	58	M	16	U
3	School A	56	M	15	U
4	School A	56	M	15	U
5	School B	56	F	18	R
6	School A	55	M	15	U
7	School A	55	F	18	U
8	School A	54	M	16	R
9	School A	54	F	17	U
10	School A	54	M	16	U
11	School A	54	F	17	R
12	School A	53	M	15	U
13	School A	53	F	18	U
14	School A	53	M	20	U
15	School A	52	M	15	U
16	School A	52	M	16	U
17	School A	52	F	17	U
18	School A	51	F	15	U
19	School A	51	M	18	R
20	School A	50	M	15	U

School	Number of students	Mean
School A	349	32.21
School B	46	30.72
Overall	395	?

Method 1 : $\frac{32.21 + 30.72}{2} = 31.47$ (2 d. p) 

Method 2 : $\frac{349}{395} \times 32.21 + \frac{46}{395} \times 30.72$

↓
Weighted average = 32.04 (2 d. p) 

Source: P. Cortez & A. Silva. Using Data Mining to Predict Secondary School Student Performance. In A. Brito and J. Teixeira Eds., Proceedings of 5th FUTURE BUSINESS TECHNOLOGY CONFERENCE (FUBUTEC 2008) pp. 5-12, Porto, Portugal, April, 2008, EUROSIS, ISBN 978-9077381-39-7

Clinical trials : Proportion as a form of mean

Imagine we want to investigate the effectiveness of a new drug for treating asthma attacks compared to an existing drug.

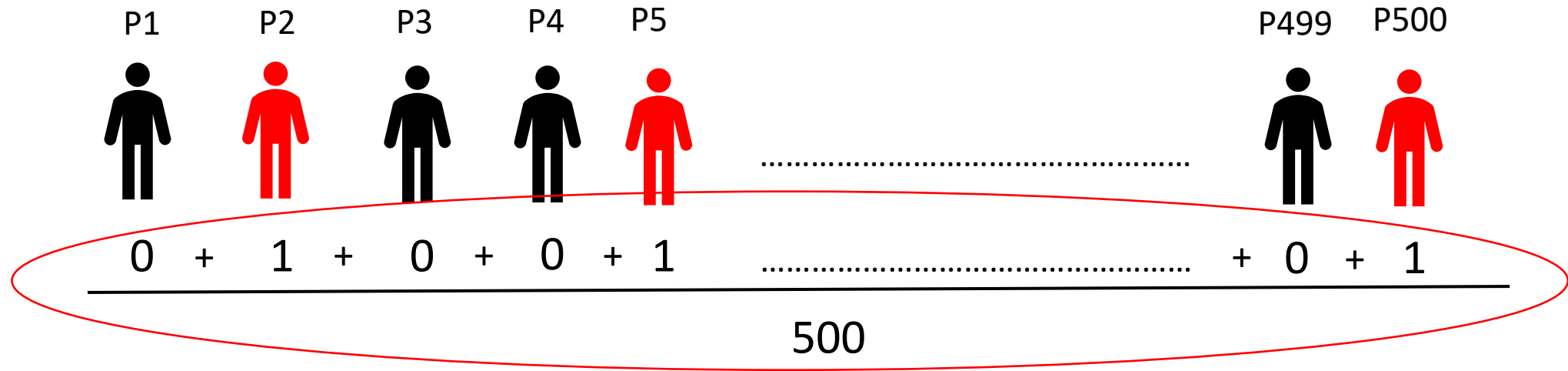
	New drug	Existing drug
Number of patients	500	1000
Number of Asthma attacks	200	300

Can we say that the new drug is more effective since there are fewer asthma attacks?

Rate(Asthma attacks for New drug) = 0.4 > Rate(Asthma attacks for Existing drug) = 0.3

So how is proportion a mean?

Suppose we lined up the 500 people who received the new drug in a line.



Moral of the story : Proportion is the mean for a numerical variable that takes **only 2 values**, namely 0 and 1.

Summary

So far we have looked at

- What is a variable and the nature of independent and dependent variables
- How to recognise numerical and categorical variables
- The need to classify variables
- The role summary statistics play in data analysis
- Definition and properties of the mean
- Overall means versus subgroup means
- How proportion is nothing but a special case of the mean.



Variables and Summary Statistics

Standard deviations, Medians,
Interquartile ranges

Mr Sudarshan NARASIMHAN

Overview

- Standard deviation, medians and interquartile ranges.
 - Formula for sample variance and standard deviation
 - Properties of standard deviations
 - Understanding EDA, means and standard deviations through Palmer Penguins dataset.
 - Definition and properties of medians
 - Quartiles and Interquartile range
 - Mode

Sample variance and standard deviation

Recall that the mean doesn't tell us anything significant about how data is distributed which also includes the spread of the data points.

The **standard deviation** is one way (*yes there are other ways*) of quantifying the “spread” of the data **about the mean**. The formula is derived via the **variance**.

$$\text{Sample Variance} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1} \quad s_x = \sqrt{\text{Variance}}$$

↓

Notation for standard deviation of x

where n denotes the number of data points and x_1 to x_n denotes the values of the numerical variable x in the data set. We assume this is sample data.

Getting an intuition behind the formula

Consider a small data-set {10, 10, 10, 10 10}

We can clearly see that there is no spread in this case. Does the formula tell us the same thing?

$$\text{Sample Variance} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1}$$

$$\text{Sample Variance} = \frac{(10 - 10)^2 + (10 - 10)^2 + \dots + (10 - 10)^2}{4} = 0$$

$$s_x = 0$$

More intuition behind the formula

Question : Why can't we do the following to quantify spread?

- Take the difference between each value and the mean
- Add up the differences to get the “total spread”
- Divide by the number of points to get an “average spread”



What's
wrong with
this?

Let's try applying that idea to a simple dataset {5, 10, 15, 20, 25}

The mean is 15.

The difference between each point and the mean gives us {-10, -5, 0, 5, 10}

What happens when you add up these differences to calculate the “total spread”?

0

Explicit computation of standard deviation

Consider a simple sample data set of just 3 points. Let the points be $\{1, 4, 7\}$ and suppose we wish to find the standard deviation of this data set. In this case $n = 3$.

Step 1: Find the average value of the data set.

In this case the average is $\frac{1+4+7}{3} = 4$.

Step 2 : Subtract the average value from each of your data points and square the answer.

We get $(1 - 4)^2 = 9$, $(4 - 4)^2 = 0$, $(7 - 4)^2 = 9$.

Explicit computation of standard deviation

Step 3: Add up the values in Step 2 and divide by $n - 1$ which in this case is 2.

In this case the variance is $\frac{9+9}{2} = 9$.

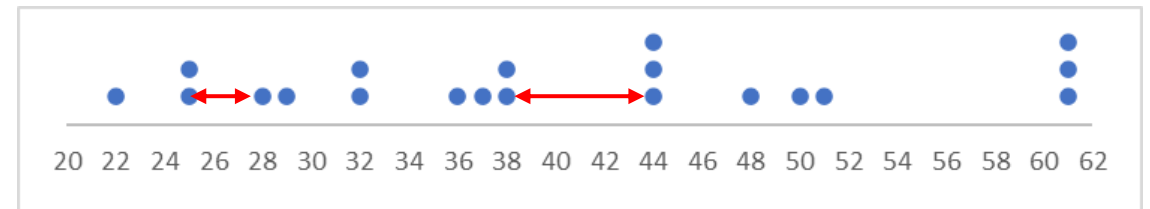
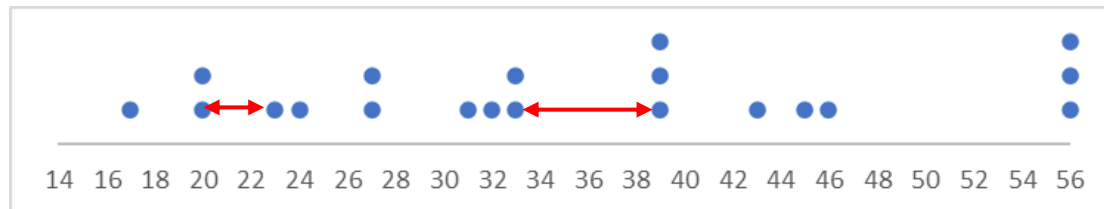
Step 4 : Square root the answer to get the standard deviation

$$s_x = \sqrt{9} = 3$$

Properties of the standard deviation

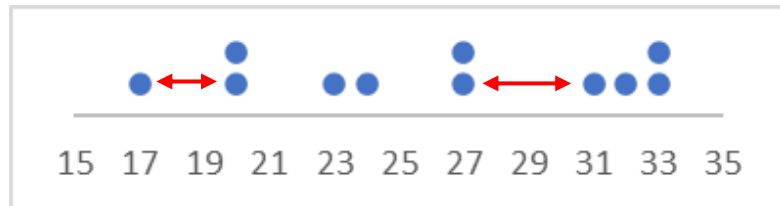
The standard deviation is always non-negative, (i.e it is either 0 or a positive number) with same units (if any) as the numerical variable

Adding a constant value, c (positive or negative) to all the data points **does not change the standard deviation.**

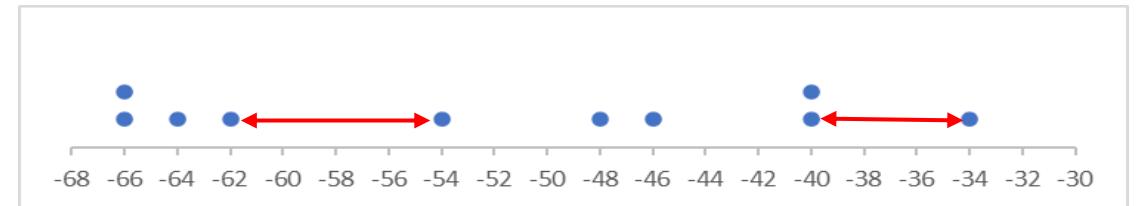


Properties of the standard deviation

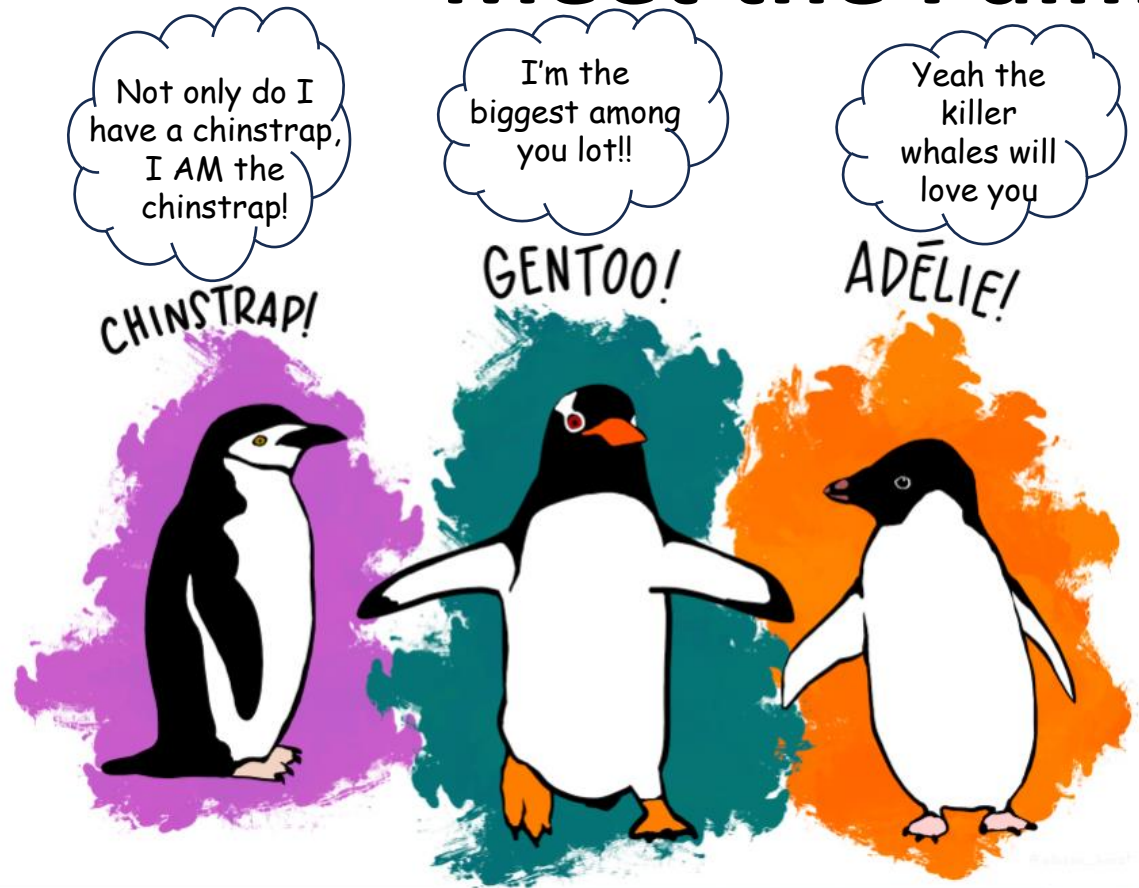
Multiplying all the data points by a constant value c results in the standard deviation being multiplied by $|c|$ where $|c|$ is the absolute value of c .



Multiply
by -2



Meet the Palmer penguins

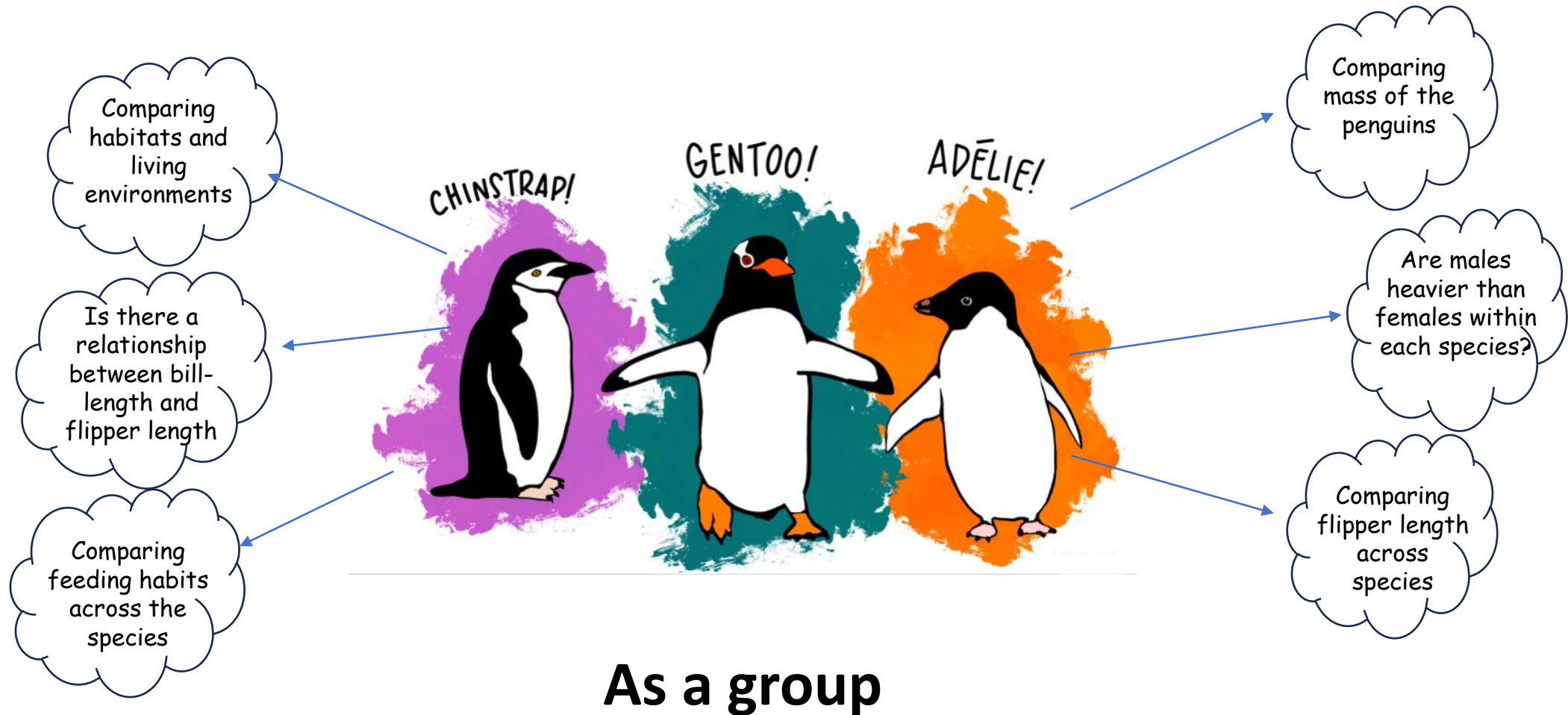


Source: allison_horst, *Meet the Palmer penguins*, 2022, Github. In the Public Domain.
<https://github.com/allisonhorst/palmerpenguins> CCO

species	island	bill_length_mm	bill_depth_mm
Chinstrap	Dream	46.9	16.6
Adelie	Biscoe	36.5	16.6
Adelie	Biscoe	36.4	17.1
Adelie	Biscoe	34.5	18.1
Adelie	Dream	33.1	16.1
Adelie	Torgersen	38.6	17
Chinstrap	Dream	43.2	16.6
Adelie	Biscoe	37.9	18.6
Adelie	Dream	37.5	18.9
Adelie	Dream	37	16.9
Adelie	Dream	37.3	16.8
Adelie	Torgersen	35.9	16.6
Adelie	Torgersen	35.2	15.9
Adelie	Torgersen	39	17.1
Adelie	Dream	32.1	15.5
Adelie	Biscoe	37.7	16
Adelie	Dream	36	18.5
Adelie	Biscoe	37.9	18.6
Adelie	Dream	36.5	18
Adelie	Biscoe	35.7	16.9

Source: [Dr. Kristen Gorman](#) and the [Palmer Station, Antarctica LTER](#).
 title/description. Attach copyright notice.

How similar are these penguins?



Comparing masses amongst the species

The standard deviation only measures an “average” spread around the mean

No

Does this mean the heaviest penguin has a mass of $4201\text{g} + 802.0\text{g} = 5003\text{g}$?

	Mean mass	Standard deviation of mass
Chinstrap	3733g	384.3g
Adelie	3701g	458.6g
Gentoo	5076g	504.1g
Overall	4201g	802.0g

A glimpse into EDA

Why is it that the Adelie and Chinstrap have almost the same mass on average but yet the standard deviation for the Adelie species is higher? Could it be due to

- Gender?
- Age? (not found in the data set)
- Location?

	Mean mass	Standard deviation of mass
Chinstrap	3733g	384.3g
Adelie	3701g	458.6g
Gentoo	5076g	504.1g
Overall	4201g	800.8g

Comparing the spread b/w variables

Let's focus on the Adelie penguin species to define a notion that helps us quantify the degree of spread *relative to the mean*.

SD of bill length	SD of flipper length
2.65mm	6.52mm
Mean bill length	Mean flipper length
38.8mm	190.0mm
Coefficient of variation	Coefficient of variation
0.07	0.03

The **coefficient of variation** is a way of quantifying the degree of spread, *relative to the mean*.

$$\text{coefficient of variation} = \frac{s_x}{\bar{x}} \text{ where } \bar{x} \neq 0$$

Median

The **median** of a numerical variable in a data-set is the middle value of the variable *after arranging* the values of the data-set in *ascending/descending order*.

Case	Age	Gender	Race	Days to Recover
1	66	M	Chinese	26
2	53	F	Chinese	14
3	37	M	Chinese	27
4	36	M	Chinese	17
5	56	F	Indian	21
6	56	M	Malay	23
7	35	M	Malay	7
8	56	F	Chinese	20
9	56	M	Chinese	25
10	56	M	Chinese	10
11	31	F	Others	11
12	37	F	Indian	13

Case	Age	Gender	Race	Days to Recover
7	35	M	Malay	7
10	56	M	Chinese	10
11	31	F	Others	11
12	37	F	Indian	13
2	53	F	Chinese	14
4	36	M	Chinese	17
8	56	F	Chinese	20
5	56	F	Indian	21
6	56	M	Malay	23
9	56	M	Chinese	25
1	66	M	Chinese	26
3	37	M	Chinese	27

Median
number
of Days To
Recover?

$$\left. \begin{array}{l} 17 \\ 20 \end{array} \right\} \frac{(17+20)}{2} = 18.5$$

Properties of the median

- Adding a constant value (positive or negative) to all the data points changes the median by that constant value.
- Multiplying all the data points by a constant value c results in the median being multiplied by c .

Overall vs subgroup medians

	A	B	C	D	E
1	school ▾	Total ▾	sex ▾	age ▾	location ▾
2	School A	58	M	16	U
3	School A	56	M	15	U
4	School A	56	M	15	U
5	School B	56	F	18	R
6	School A	55	M	15	U
7	School A	55	F	18	U
8	School A	54	M	16	R
9	School A	54	F	17	U
10	School A	54	M	16	U
11	School A	54	F	17	R
12	School A	53	M	15	U
13	School A	53	F	18	U
14	School A	53	M	20	U
15	School A	52	M	15	U
16	School A	52	M	16	U
17	School A	52	F	17	U
18	School A	51	F	15	U
19	School A	51	M	18	R
20	School A	50	M	15	U

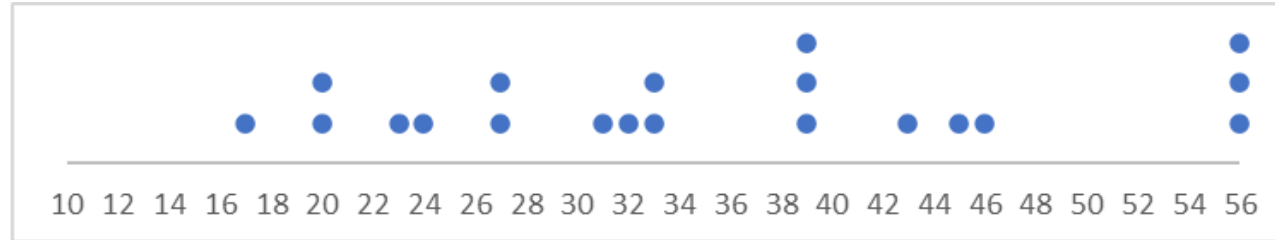
School	Number of students	Median
School A	349	32
School B	46	30.5
Overall	395	32

The overall median will be in between (possibly including) the subgroup medians.

Question : Will it be closer to the median of the larger subgroup all the time?

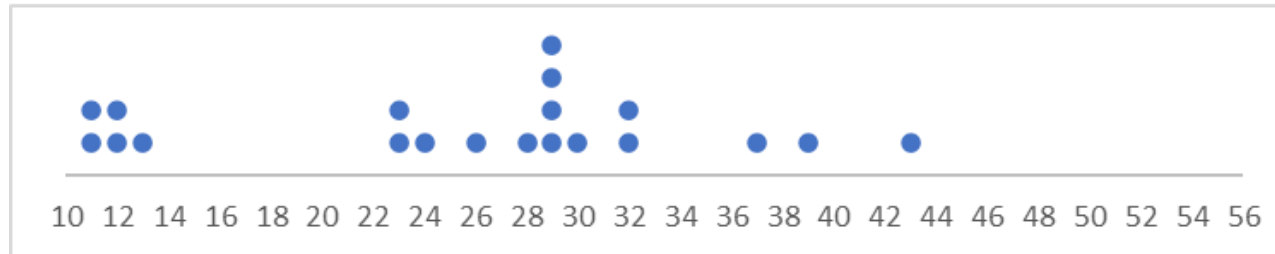
Overall vs subgroup medians

Students in Class A



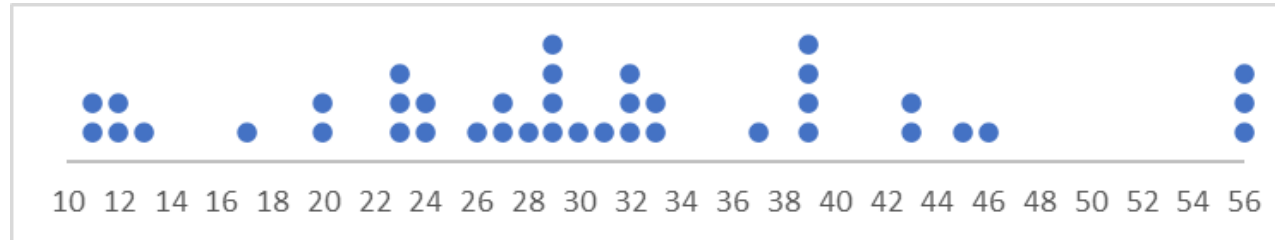
Median = 33

Students in Class B



Median = 28.5

Class A + Class B

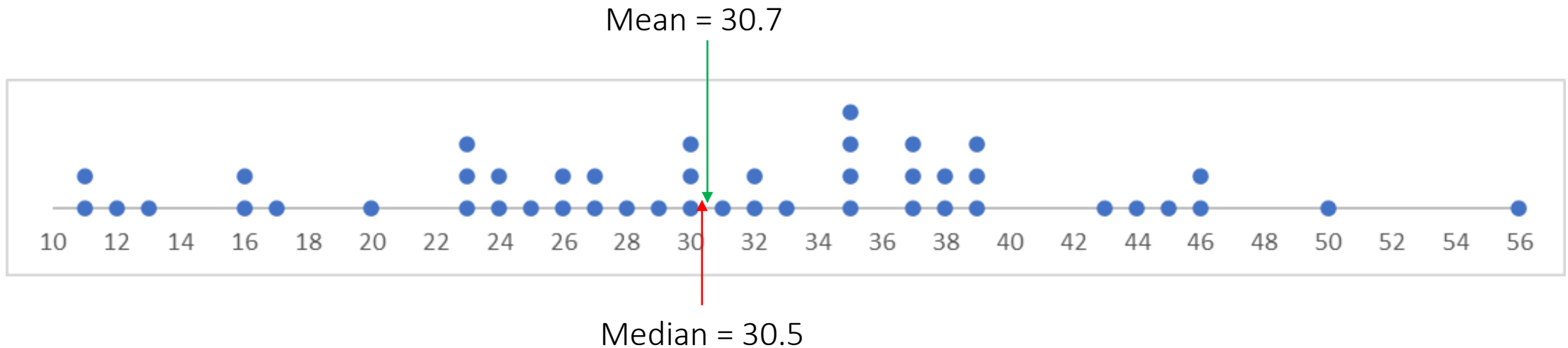


Median = 29

The relationship between means and medians

Question : When are means and medians the same?

Let's try to answer this using a dot plot of the 46 students from School B



When distributions of points are roughly symmetric, the mean and median will be quite close to each other.

Quartiles and InterQuartileRange

The **first quartile** usually denoted by Q_1 is the 25th percentile of the data values and the **third quartile**, usually denoted by Q_3 is the 75th percentile of the data-values.

(The 25th percentile, is a value such that 25% of the data is either equal to or less than this value. Same idea for the 75th percentile)

The **interquartile range** is the difference between the third and the first quartile.

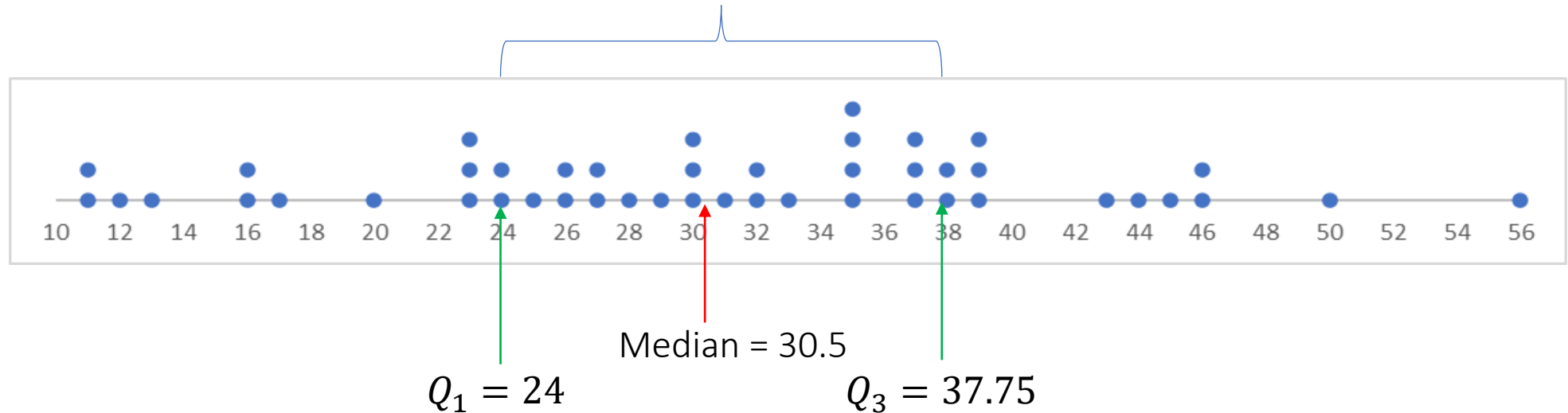
$$IQR = Q_3 - Q_1$$

It gives us another way of quantifying the spread of the data.

Explicit computation of IQR

Let's go back to the same 46 students from School B and explicitly work out the IQR

$$IQR = 37.75 - 24 = 13.75$$



11, 11, 12, 13, 16, 16, 17, 20, 23, 23, 23, 24, 24, 25, 26, 26, 27, 27, 28, 29, 30, 30, 30.

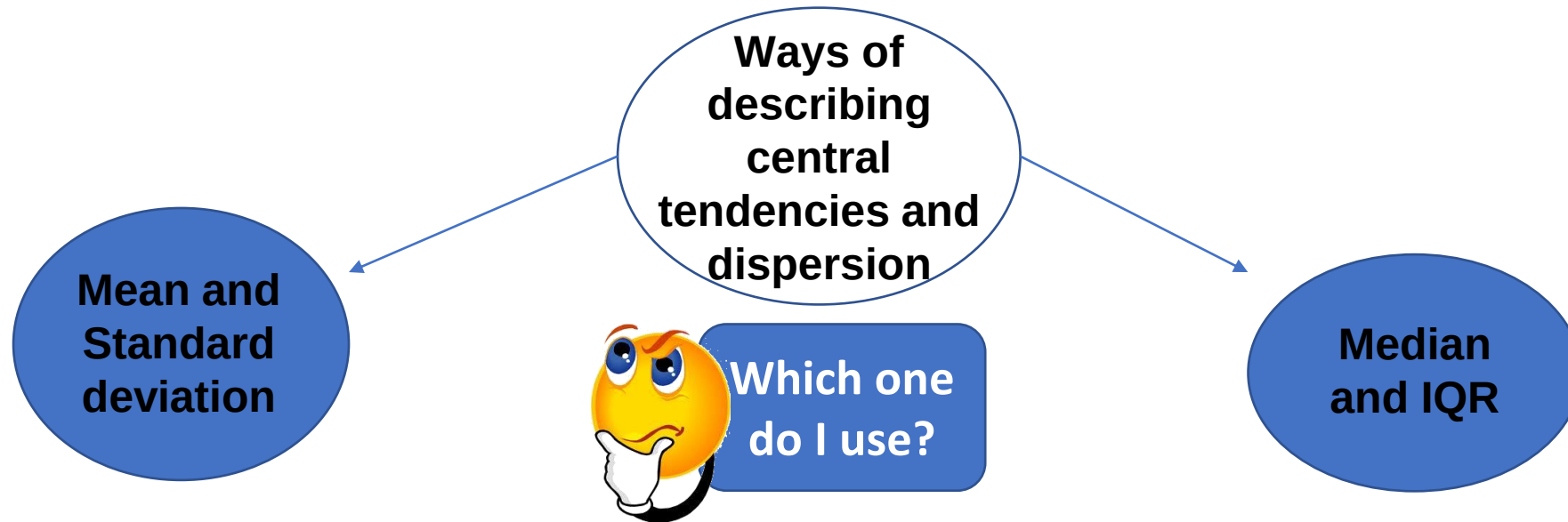
Similarities b/w IQR and Standard Deviation

The IQR is always non-negative and this follows from the fact that Q_3 is at least as large as Q_1 .

Adding a constant value, c (positive or negative) to all the data points **does not change the IQR**.

Multiplying all the data points by a constant value c results in the IQR being multiplied by $|c|$.

Using Summary statistics appropriately



Mode

Mode of a variable is the value of the variable that appears the most frequently.

Case	Age	Gender	Race	Days to Recover
1	66	M	Chinese	26
2	53	F	Chinese	14
3	37	M	Chinese	27
4	36	M	Chinese	17
5	56	F	Indian	21
6	56	M	Malay	23
7	35	M	Malay	7
8	56	F	Chinese	20
9	56	M	Chinese	25
10	56	M	Chinese	10
11	31	F	Others	11
12	37	F	Indian	13

Mode for Gender?

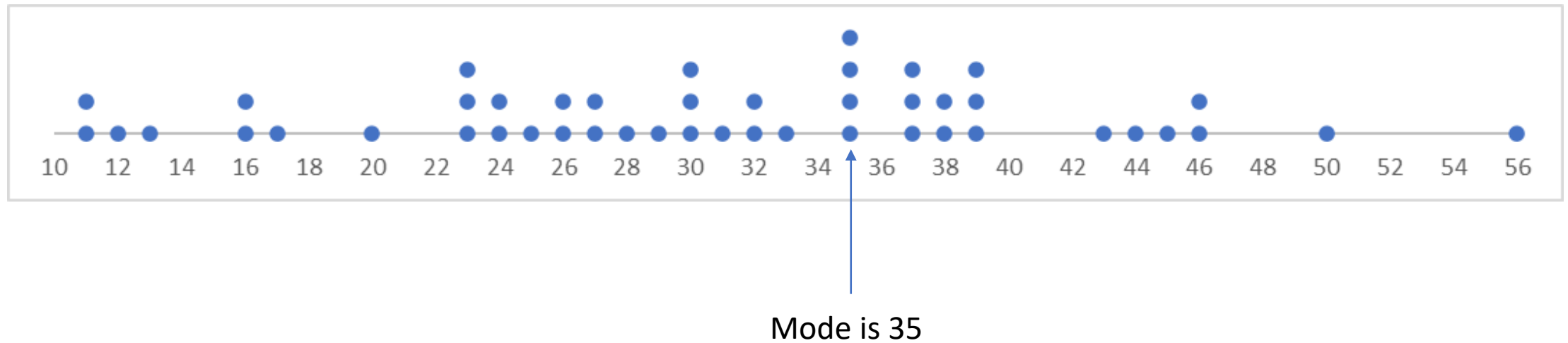
Male

Mode for Age?

56

Mode as the “peak” of a distribution

For the final time, let's go back to that dot-plot of the 46 students from School B and see what the mode is telling us.



Summary

- Understanding how the formula for standard deviation quantifies the spread of the data
- Properties of standard deviations
- A glimpse of how EDA looks like using the Palmer Penguins data set

Summary

- Understanding median as the middle value of a data set after ordering
- Understanding how the interquartile range is another measure of spread
- Mode and its interpretation as a peak of the distribution



Study Designs I

Mr Samuel YEUN

Overview

- Experimental Studies
 - Treatment and Control Groups
 - Random Assignment
 - Blinding
- Observational Studies
- Experimental vs Observational Studies



Study Designs I

Experimental Studies

Mr Samuel YEUN

Recall

Type of Research Questions

Make an estimate about the population

Test a claim about the population

Compare two sub-populations

Investigate a relationship
between two variables in the population

Recall

Type of Research Questions

Make an estimate about the population

Test a claim about the population

Compare two sub-populations

Investigate a relationship
between two variables in the population



Does
drinking coffee
help students
pass the math exam?

Types of Study Designs

Experimental

Observational

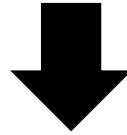


Experimental Studies

An experiment intentionally manipulates one variable in an attempt to cause an effect on another variable.

The primary goal of an experiment is to provide evidence for a cause-and-effect relationship between two variables.

Independent Variable



Dependent Variable

Experimental Studies

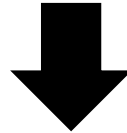
An experiment intentionally manipulates one variable in an attempt to cause an effect on another variable.

The primary goal of an experiment is to provide evidence for a cause-and-effect relationship between two variables.



Independent Variable (Drinking coffee)

Help students



Dependent Variable (Pass the math exam)

Experimental Studies – Treatment and Control

Coffee	No coffee
<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>



Experimental Studies – Treatment and Control

Coffee	No coffee
<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>

**Treatment
Group**

**Control
Group**



Experimental Studies – Treatment and Control

Coffee

*Drink exactly one cup of coffee every
day for one month*

**Treatment
Group**



Experimental Studies – Treatment and Control

Coffee

Drink exactly one cup of coffee every day for one month

**Treatment
Group**

90% passed



~~Coffee helps?~~



Experimental Studies – Treatment and Control

Coffee	No coffee
<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>

**Treatment
Group**



90% passed

Maybe???

90% passed

Experimental Studies – Treatment and Control

Coffee	No coffee
<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>

**Treatment
Group**

**Control
Group**



The control group provides a baseline for comparison with the treatment group.

Experimental Studies – Treatment and Control

Antidepressant

Nicotine patch

**Treatment
Group**

**Control
Group**

The control group provides a
baseline for comparison with the
treatment group.





Study Designs I

Experimental Studies (Random Assignment)

Mr Samuel YEUN

Experimental Studies – Study Results

Coffee	No coffee
<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>

**Treatment
Group**

**Control
Group**



Experimental Studies – Study Results

	Coffee	No coffee
	<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>
Pass		
Fail		



Experimental Studies – Study Results

	Coffee	No coffee
	<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>
Pass	900	450
Fail	100	550

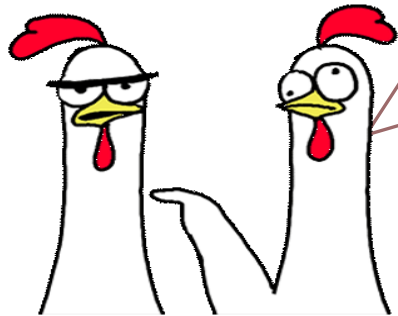


90% passed

45% passed

Experimental Studies

	Coffee	No coffee
	<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>
Pass	900	450
Fail	100	550



Coffee causes an improvement in passing the math exam!!!

Not wise?

Experimental Studies



Pass the math exam



Revision time

Experimental Studies

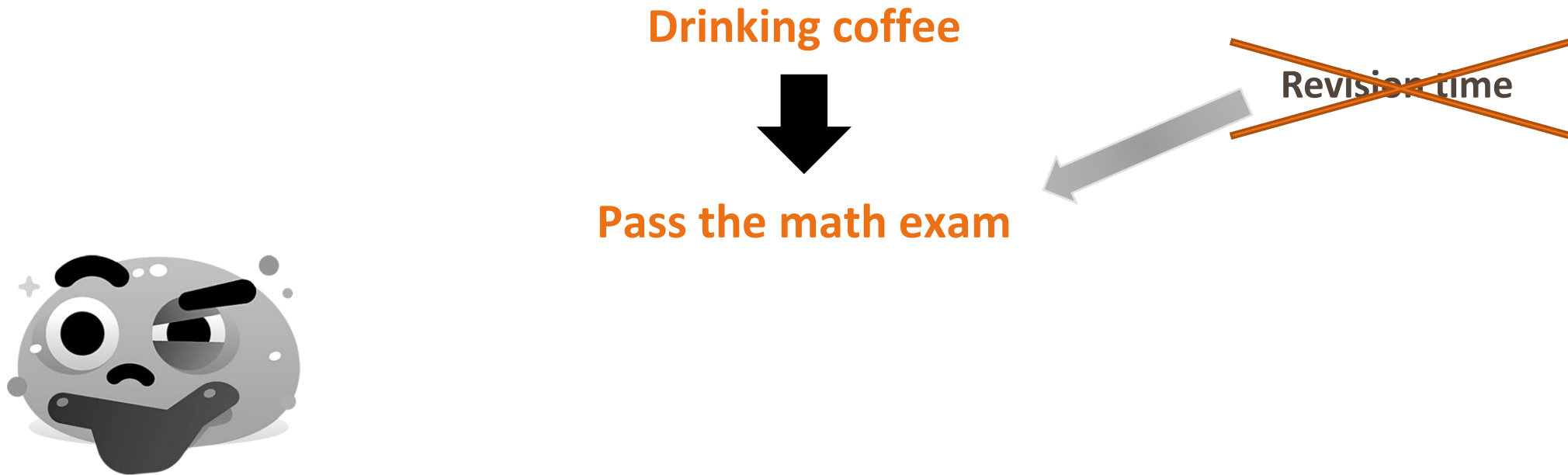
	Long Revision Time	Short Revision Time
Pass	900	450
Fail	100	550



possible?

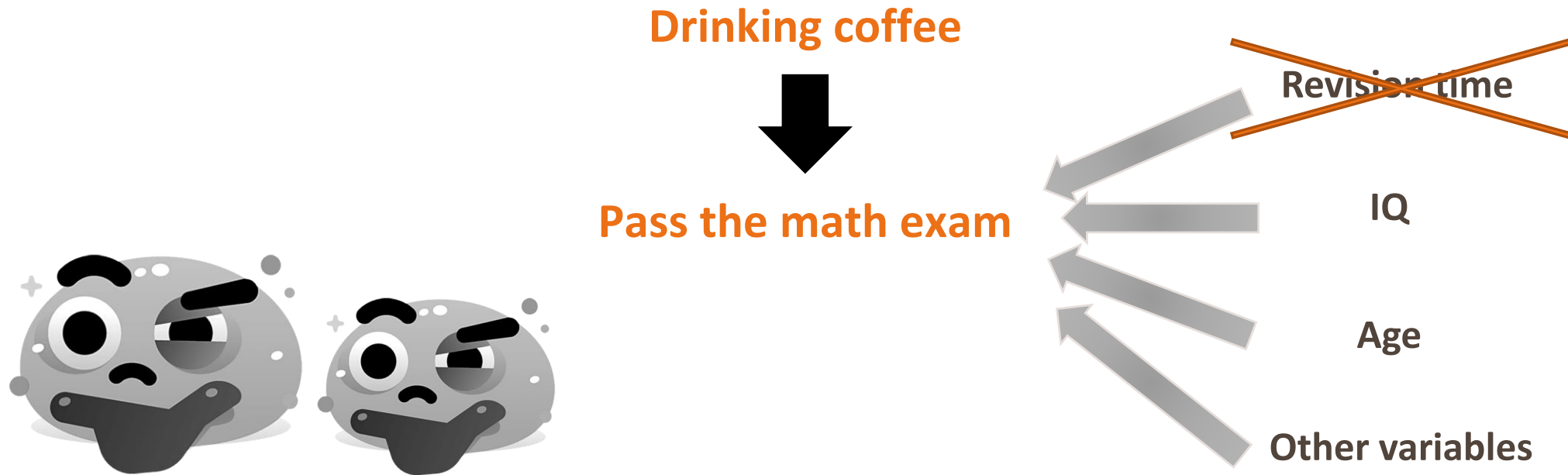
Experimental Studies

To establish a cause-and-effect relationship, we want to make sure that the independent variable is the only factor that impacts the dependent variable.



Experimental Studies

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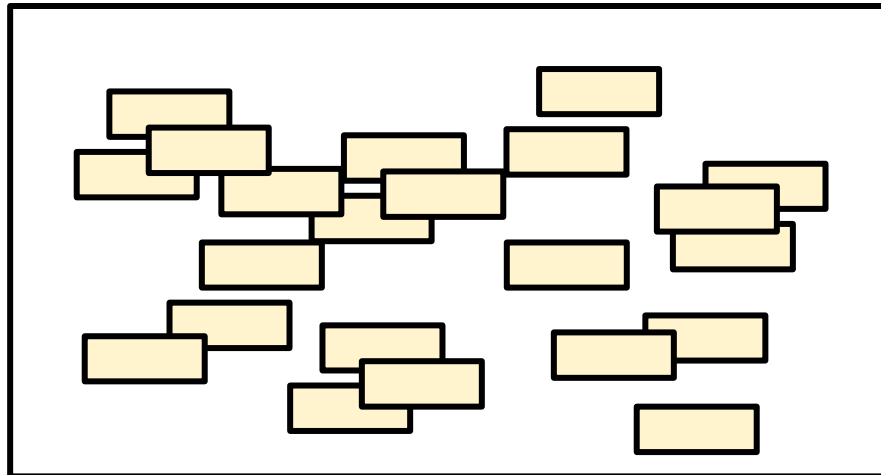
Experimental Studies

How do we account for the effects from all these other variables?

Random Assignment

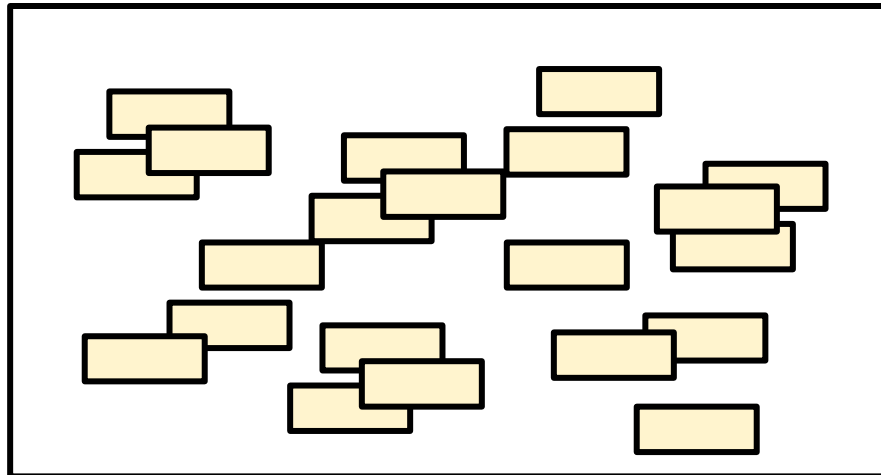
Random Assignment

Random assignment is an impartial procedure that uses chance.



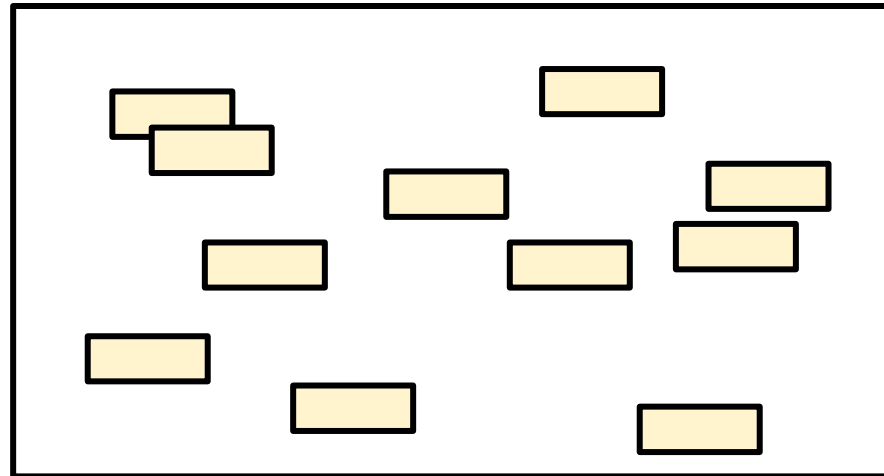
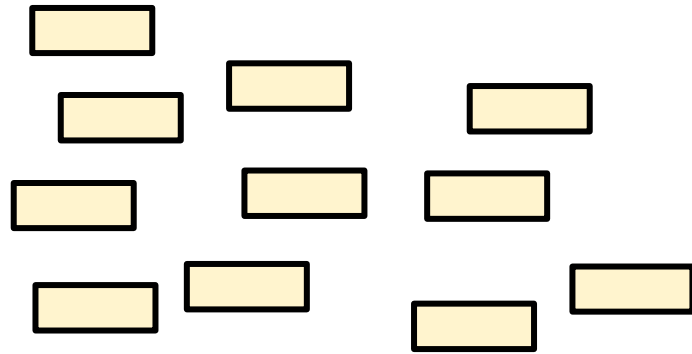
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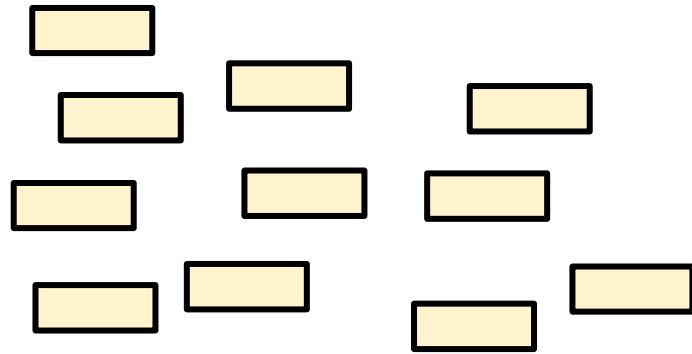
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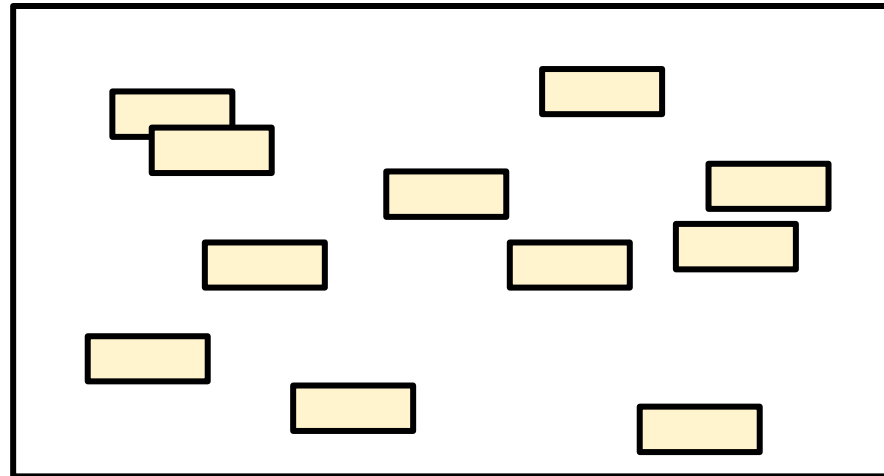


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Treatment Group

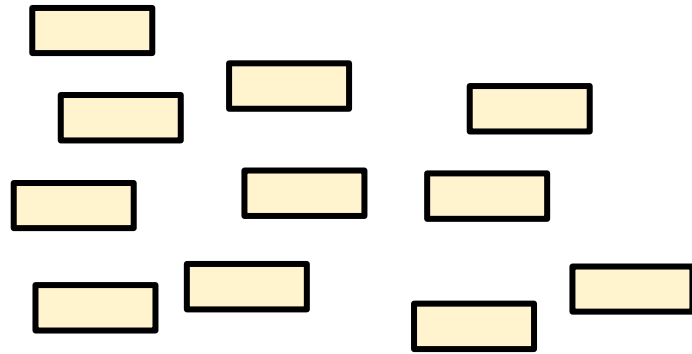


Control Group

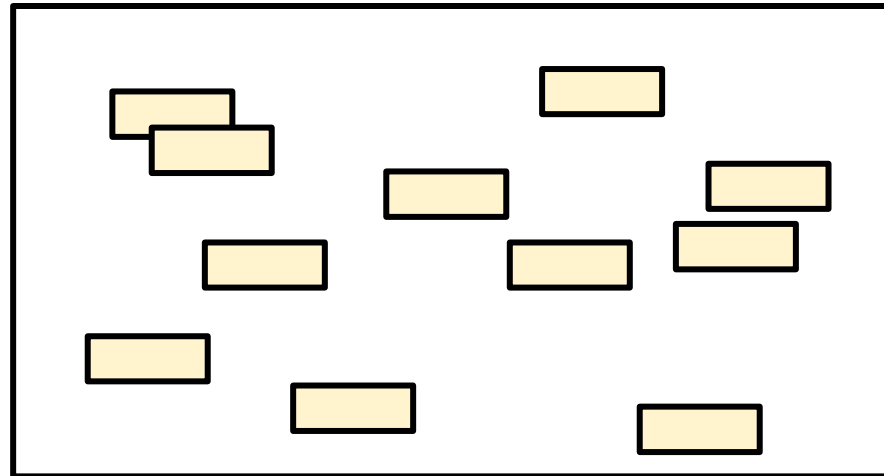
Random Assignment

Random assignment is an impartial procedure that uses chance.

If the number of subjects is large, by the laws of probability, the treatment and control groups will tend to be similar in all aspects.



Treatment Group



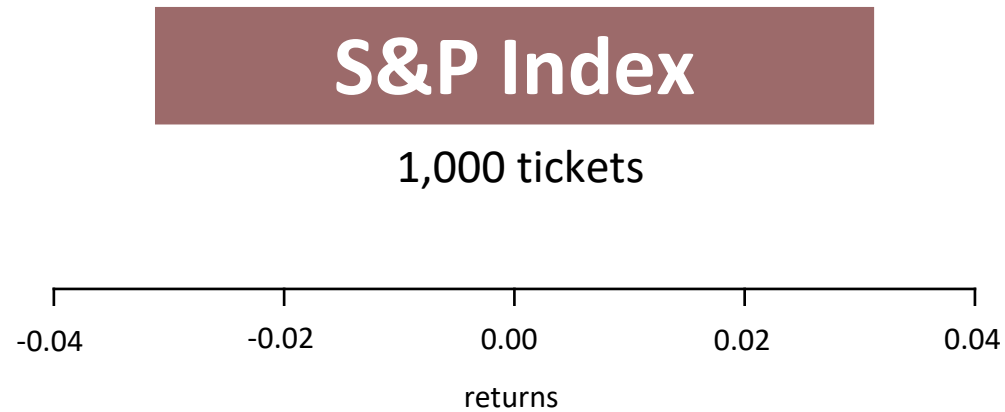
Control Group

Random Assignment

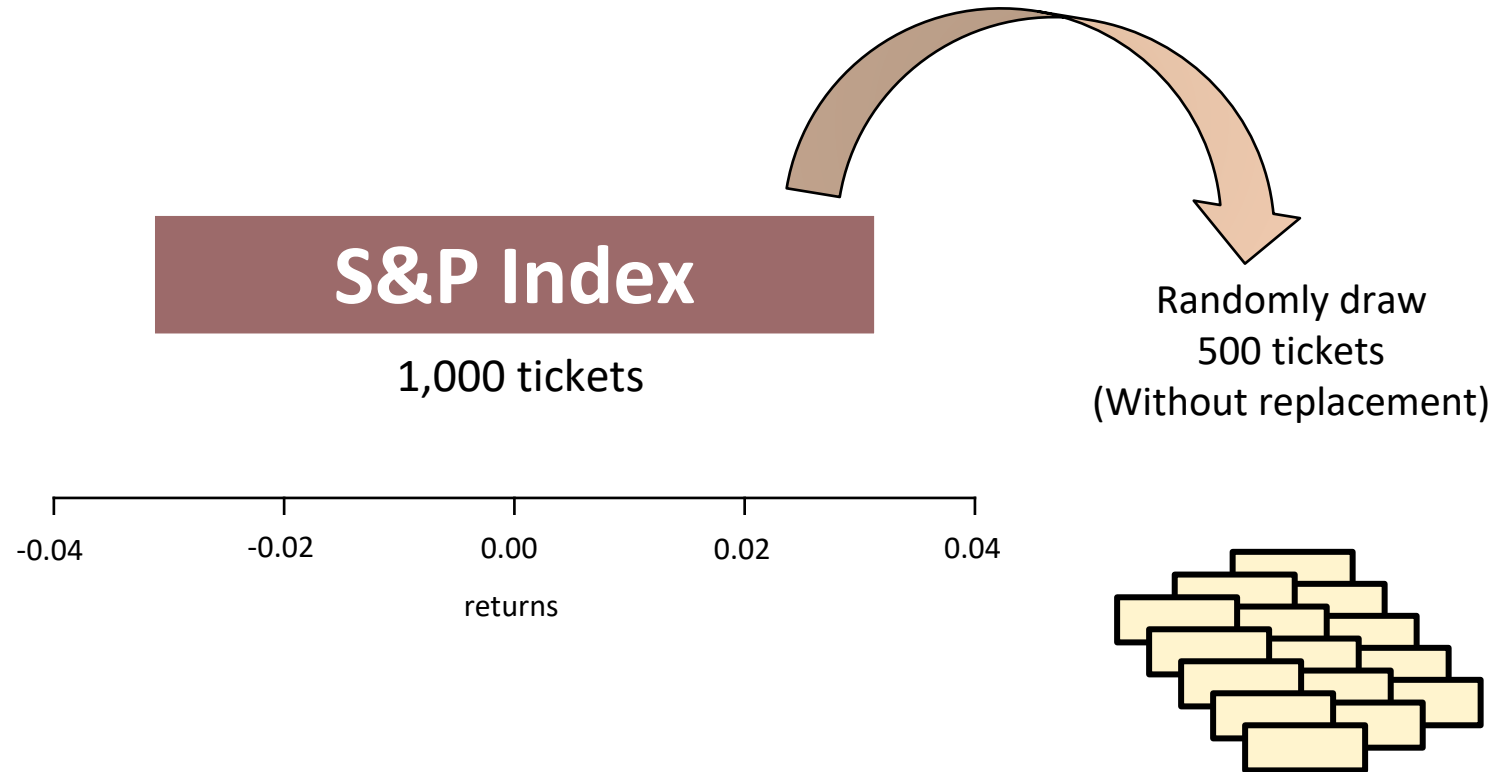
The goal of random assignment is to create similar treatment and control groups with respect to

- Revision time
- IQ
- Age
- Other variables

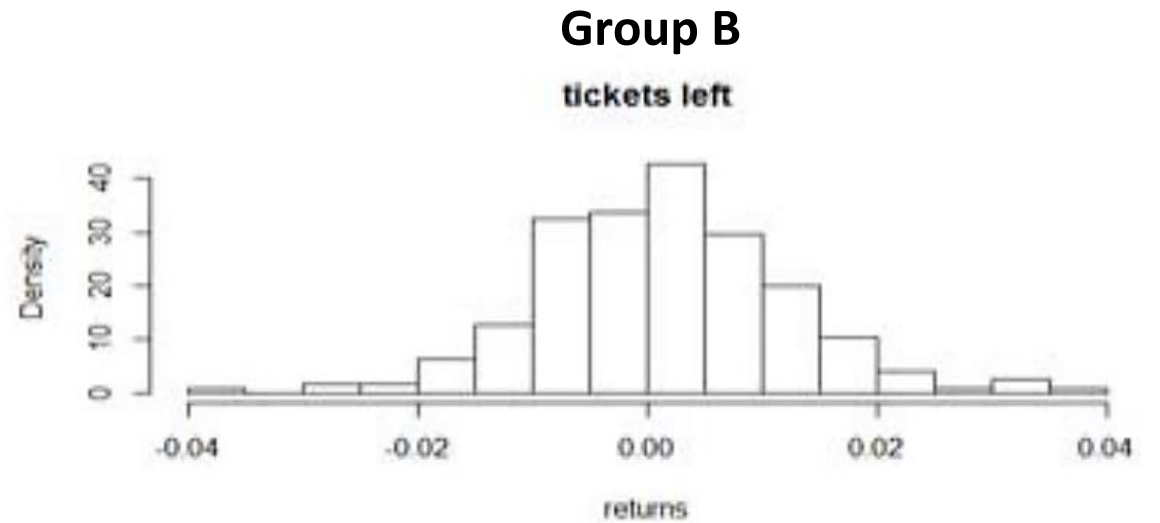
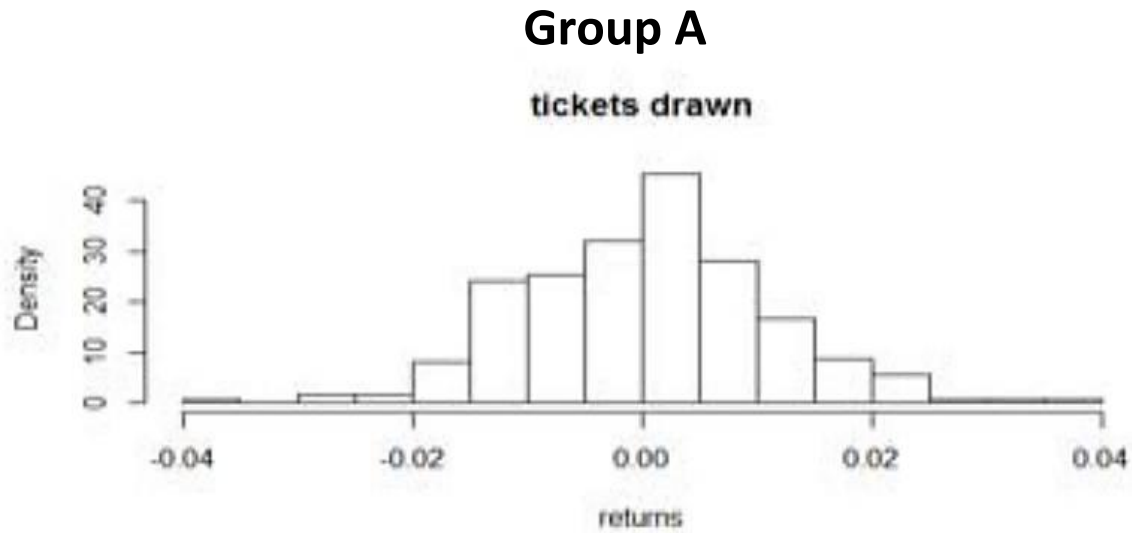
Random Assignment – S&P Example



Random Assignment – S&P Example

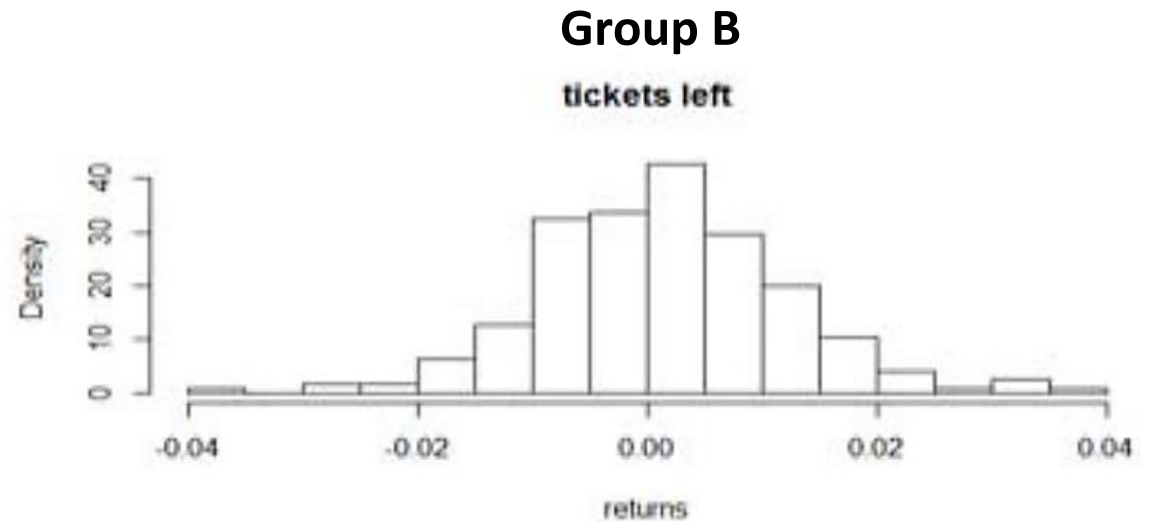
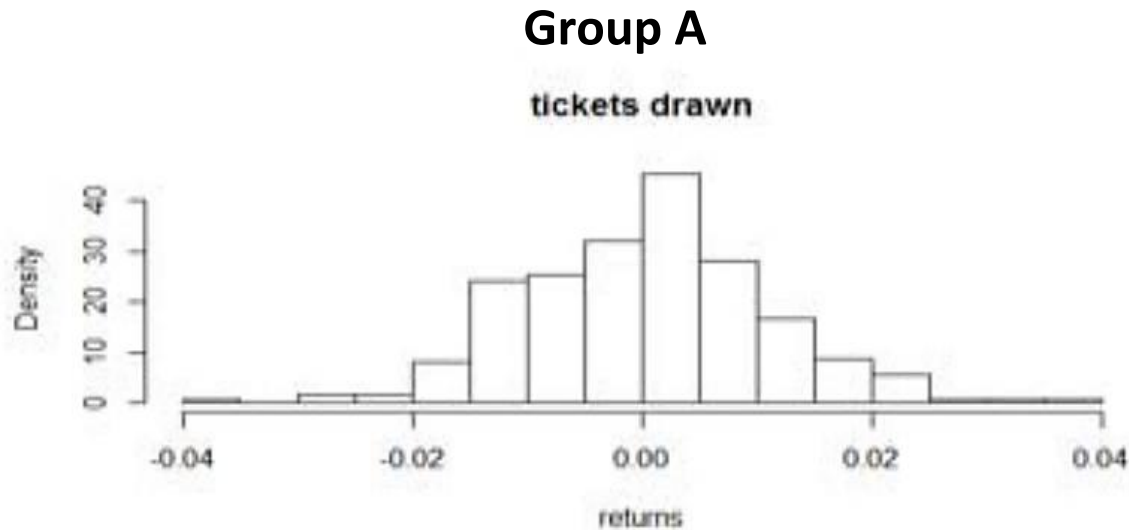


Random Assignment – S&P Example



Similar distributions

Random Assignment – S&P Example



The treatment and control groups can have different sizes.
As long as the size of the groups are quite large, then a randomised assignment tends to produce two very similar groups.

Note on the term “Random”

“Random” has a much more strict meaning related to an impartial chance mechanism.

- “Random” does not mean “Haphazard”.

Summary

- Experimental Studies
 - Treatment and Control Groups
 - Random Assignment



Study Designs

Mr Samuel YEUN

Overview

- Experimental Studies
 - Treatment and Control Groups
 - Random Assignment
 - Blinding
- Observational Studies
- Experimental vs Observational Studies



Study Designs

Experimental Studies (Blinding)

Mr Samuel YEUN

Experimental Studies

Coffee	No coffee
<i>Drink exactly one cup of coffee every day for one month</i>	<i>Not drink any coffee for one month</i>

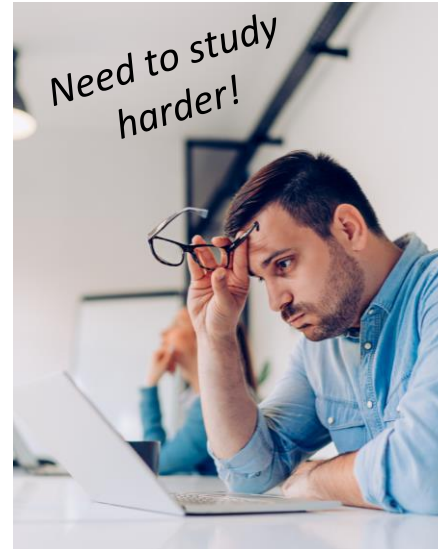
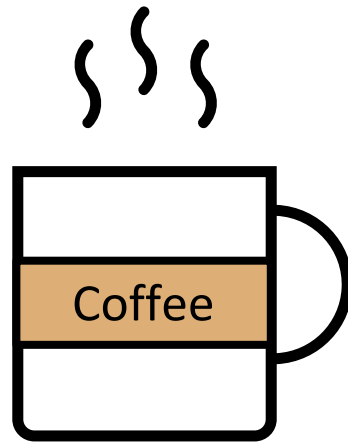
**Treatment
Group**

**Control
Group**

In some cases, leaving the control group alone may cause bias!

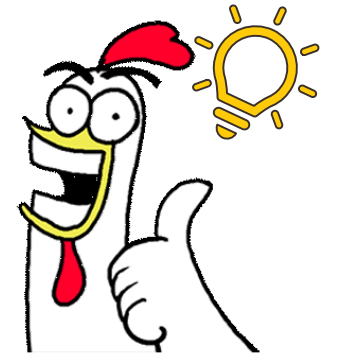
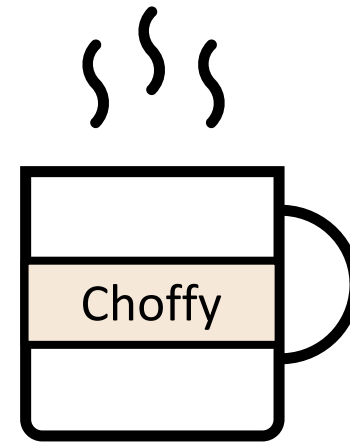
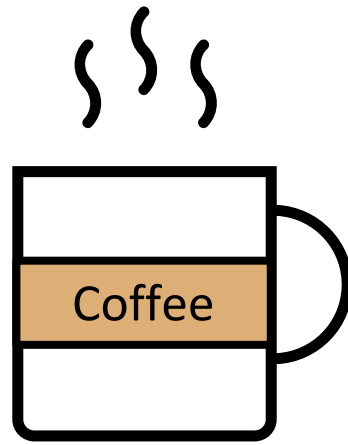
Experimental Studies

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Experimental Studies

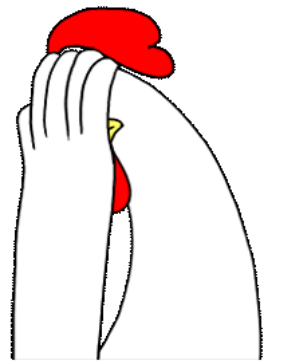
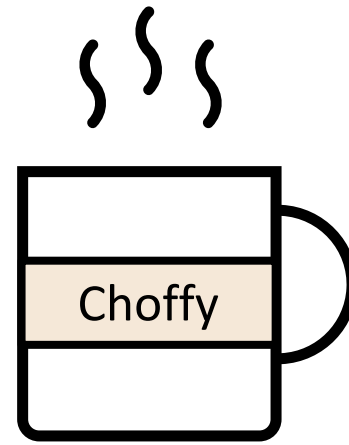
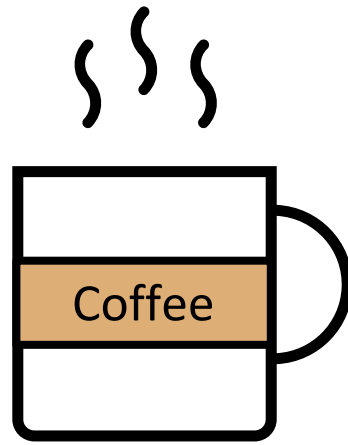
Coffee	No coffee
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Experimental Studies - Placebo

Placebo: Treatment with no active ingredients, and no effect.

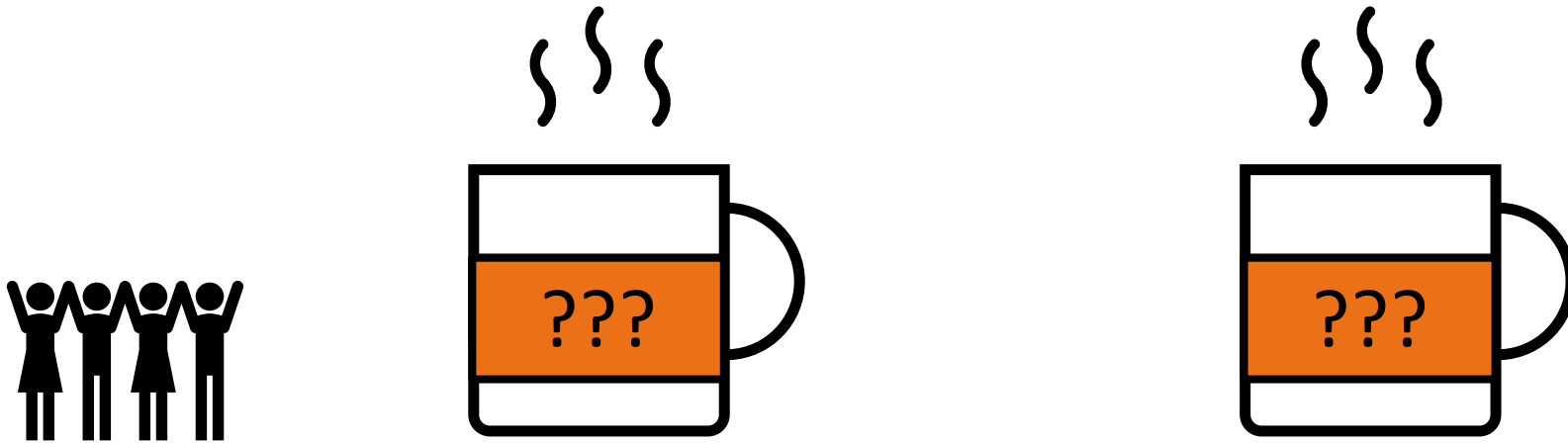
Placebo Effect: The response observed when subjects receive a placebo treatment, but still show some positive effects.



Blinding

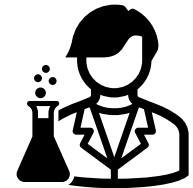
Blinded subjects do not know whether they are in the treatment or control group.

- A placebo that is very similar to the treatment can be chosen to help make the blinding effective.
- The subjects are blind to the treatment to prevent their own beliefs about the treatment from affecting the results.



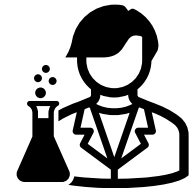
Blinding

Blinded assessors do not know whether they are assessing the treatment or control group.



Blinding

An experiment is called double-blind if both subjects and assessors are blinded about the assignment.





Study Designs

Experimental vs Observational Studies

Mr Samuel YEUN

Experimental Studies

Do **vaccinations** help reduce the **effects of the coronavirus**?

NEWS | 20 October 2020

Dozens to be deliberately infected with coronavirus in UK 'human challenge' trials

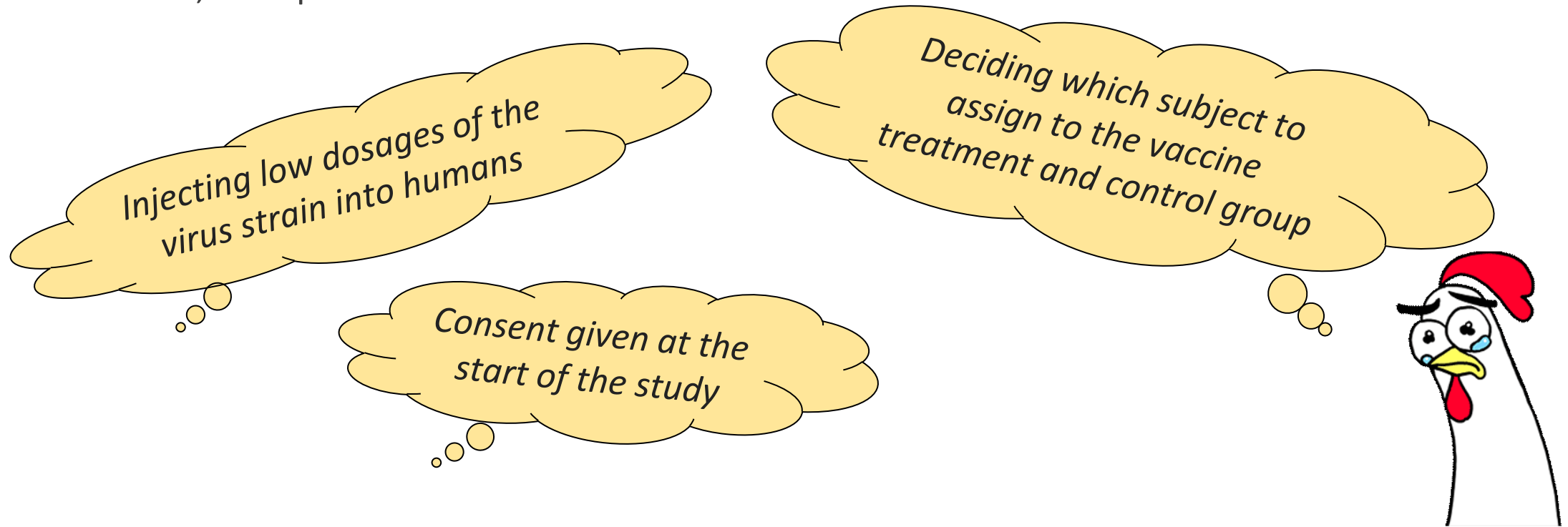
Proponents of the trials say they can be run safely and help to identify effective vaccines, but others have questioned their value.

Source: Callaway E., "Dozens to be deliberately infected with coronavirus in UK 'human challenge' trials", 2020, Nature News)

Ethical Issues

Experiments are useful in providing evidence for a cause-and-effect relationship.

However, an experiment has its issues too.



Types of Study Designs

Experimental

Observational



Observational Studies

An observational study observes individuals and measures variables of interest.

However, researchers do not attempt to directly manipulate one variable to cause an effect in another variable.

- Does not provide convincing evidence of a cause-and-effect relationship.

Observational Studies

Is **long term smoking** linked to **heart disease**?

Research

Impact of smoking and smoking cessation on cardiovascular events and mortality among older adults: meta-analysis of individual participant data from prospective cohort studies of the CHANCES consortium

Source: Mons et al., "Impact of smoking and smoking cessation on cardiovascular events and mortality among older adults: Meta-analysis of individual participant data from prospective cohort studies of the chances consortium," 2015, BMJ

Observational Studies

Is **long term smoking** linked to **heart disease**?

Note: For observational studies, while there is no actual treatment being assigned to the subjects, we still use the terms “treatment” and “control” groups in the same way as though we are dealing with an experiment.

Smokers	Non-smokers
Treatment (Exposure) Group	Control (Non-exposure) Group

Experimental vs Observational Study

Experimental Studies	Observational Studies
Assigned by researcher	Assigned by subjects themselves

Experimental vs Observational Study

Experimental Studies	Observational Studies
Assigned by researcher	Assigned by subjects themselves
Can provide evidence of a cause-and-effect relationship	Cannot provide evidence of a cause-and-effect relationship

Our study corroborates and expands evidence from previous studies in showing that smoking is a strong independent risk factor of cardiovascular events and mortality ...

Experimental vs Observational Study

Experimental Studies	Observational Studies
Assigned by researcher	Assigned by subjects themselves
Can provide evidence of a cause-and-effect relationship	Can provide evidence of 'Association'

Final note - Generalisability

If an experiment is well-designed, can we generalise the results?

Final note - Generalisability



If an experiment is well-designed, can we generalise the results?

Sampling frame?
Sampling method?
Sample size?

The design of the experiment is not the only factor we look at.

Summary

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